



國立陽明交通大學  
NATIONAL YANG MING CHIAO TUNG UNIVERSITY



# 深度學習技術的基礎與其在醫學影像上的應用

## Fundamentals and Medical Image Applications of Deep Learning

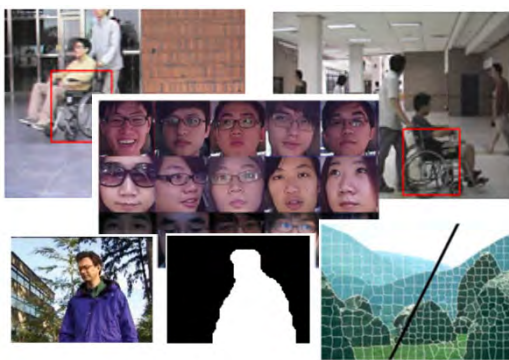
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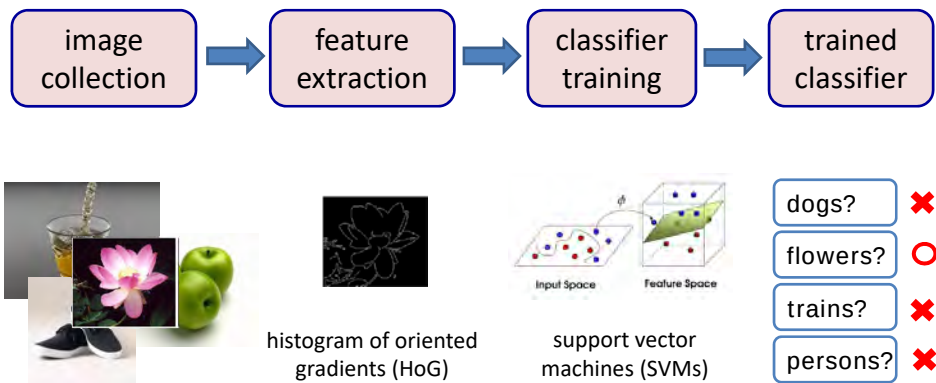
## Introduction

- What is artificial intelligence (AI)?
  - A device perceives its environment and takes actions to achieve its goals
- Applications in
  - Detection
  - Recognition
  - Segmentation



## Conventional AI Approach

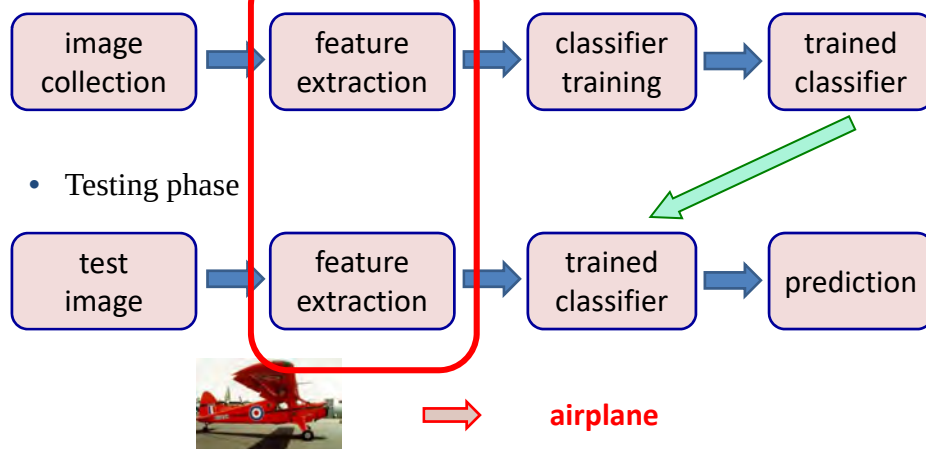
- Training phase for supervised learning



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## Conventional AI Approach

- Training phase for supervised learning



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## What Is the Most Important Feature?



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## Conventional Approaches vs. Deep Learning

- Conventional approaches
  - Fixed/engineered features + trainable classifier



Hand-crafted  
Feature Extractor

"Simple" Trainable  
Classifier

- Deep learning / End-to-end learning / Feature learning
  - Trainable features + trainable classifier



Trainable  
Feature Extractor

Trainable  
Classifier

Image Credit: Y LeCun & MA Ranzato

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## Deep Learning for Images

- Learn hierarchical representations

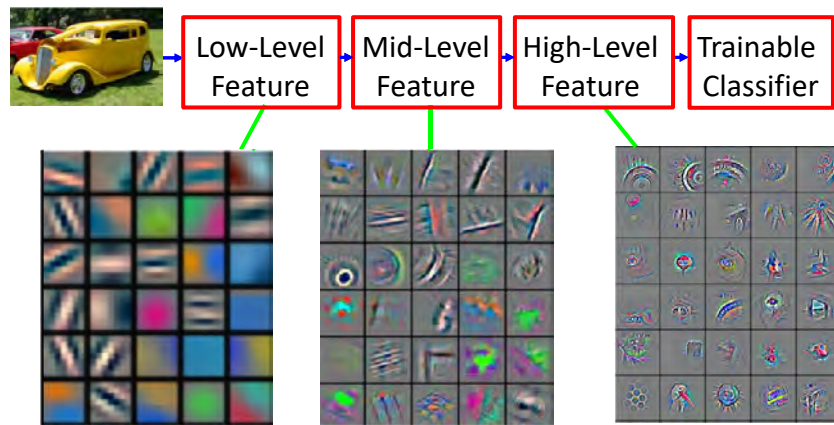
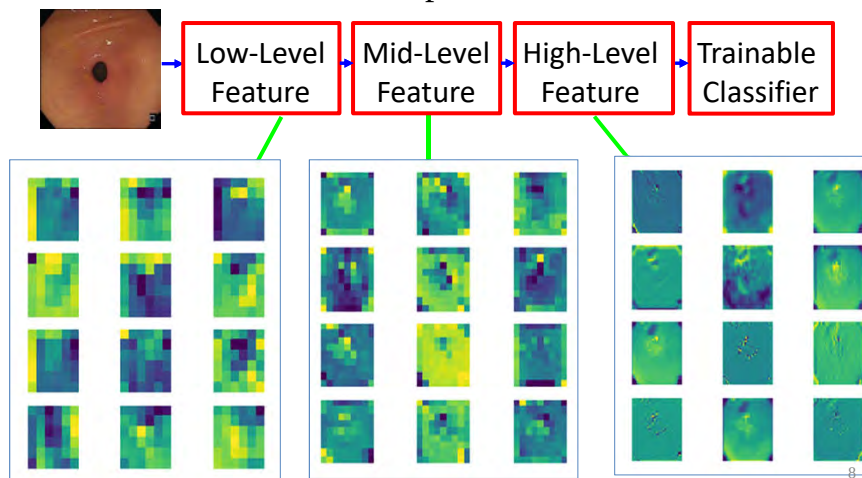


Image Credit: Y LeCun & MA Ranzato

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## Deep Learning for Medical Images ?

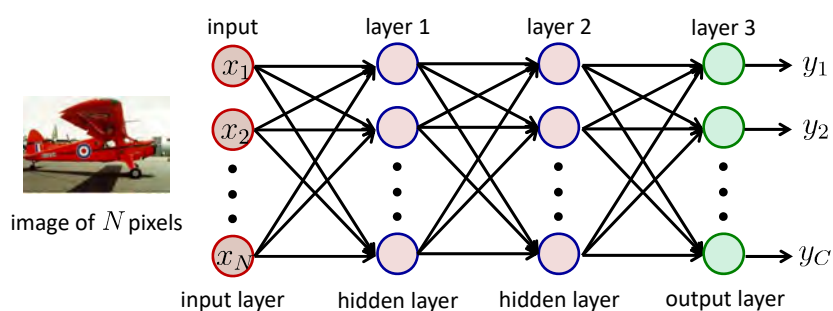
- Also learn hierarchical representations



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## Neural Networks and Neurons

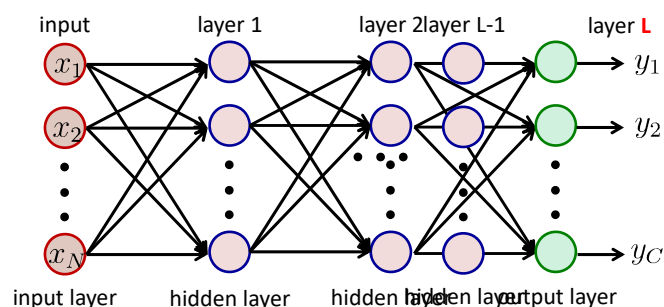
- Neural networks are presented as layers of interconnected neurons
  - Each layer of neurons takes messages from output of previous layer



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## What are Deep Neural Networks?

- DNN is a neural network with many hidden layers

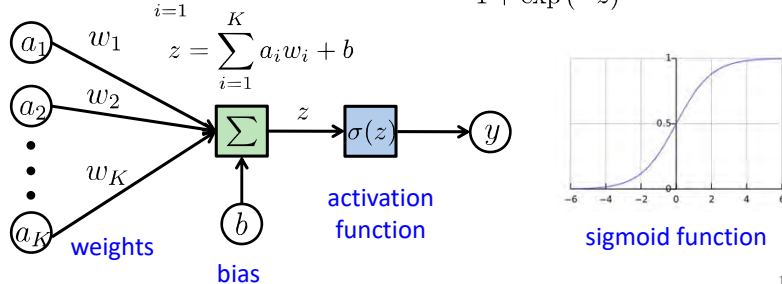


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## A Single Neuron

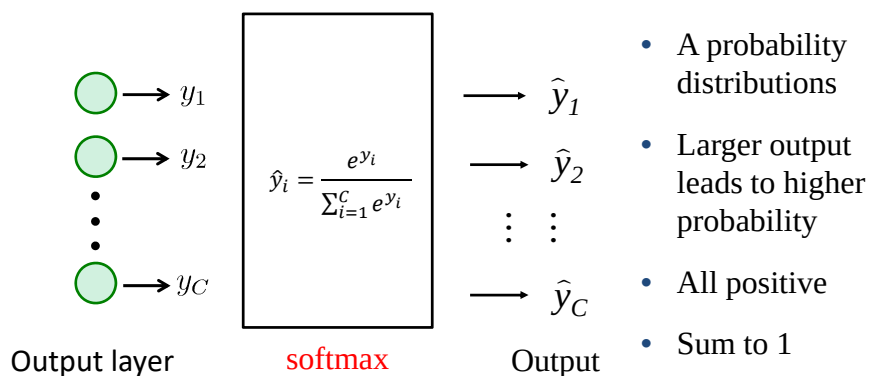
- A function  $f : R^K \mapsto R$ 
  - Map  $K$  inputs to 1 output
  - Compute the biased weighted sum
  - Apply a non-linear mapping function (activation function)

$$f((a)) = \sigma\left(\sum_{i=1}^K a_i w_i + b\right), \text{ where } \sigma(z) = \frac{1}{1 + \exp(-z)}$$



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## Softmax for Multi-class Classification



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## Training Neural Networks

- Collect a set of labeled training data  $D = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$
- Training neural networks: Finding network parameters  $\theta = \{\mathbf{w}, \mathbf{b}\}$  to minimize the loss between true training label  $\mathbf{y}_i$  and the estimated label, e.g.,

$$L(\theta) = \sum_{i=1}^N \|\mathbf{y}_i - g_{\mathbf{w}}(\mathbf{x}_i)\|^2$$

- Minimization can be done by gradient descent if  $L(\cdot)$  is differentiable with respect to  $\theta$
- **Back-propagation**: a widely used method for optimizing multi-layer neural networks

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## Stochastic Gradient Descent

- **Stochastic gradient descent** (or called sequential gradient descent) has proved useful in practice when training neural networks on a large data set
- The error function needs to comprise a sum of terms, one for each data point, i.e.,

$$E(\mathbf{w}) = \sum_{n=1}^N E_n(\mathbf{w})$$

- **Sum-of-squares error** for regression

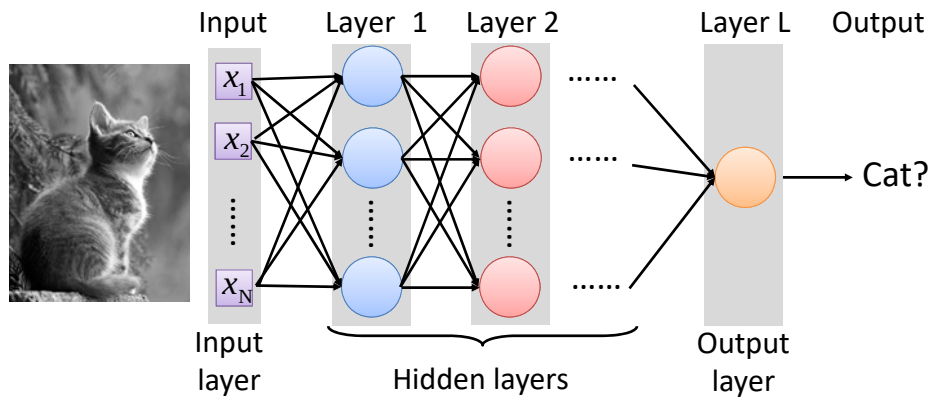
$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(\mathbf{x}_n, \mathbf{w}) - t_n\}^2$$

- **Cross-entropy error** for classification

$$E(\mathbf{w}) = - \sum_{n=1}^N \{t_n \ln y_n + (1 - t_n) \ln(1 - y_n)\}$$

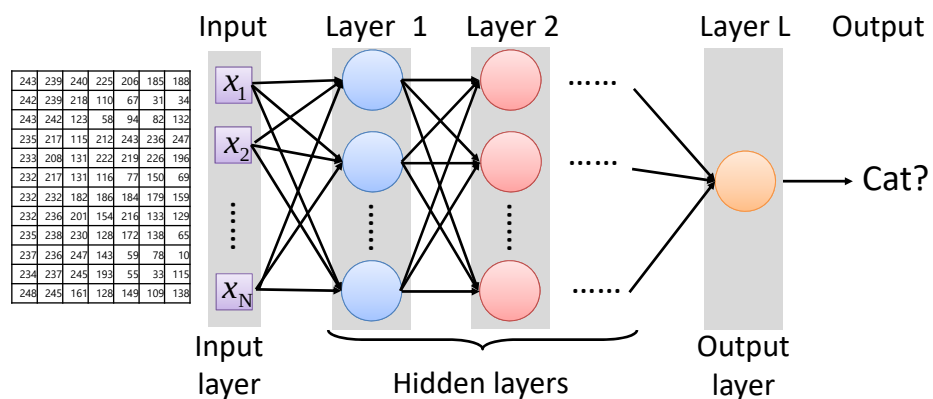
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## Deep Learning for Object Recognition



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## Deep Learning for Object Recognition

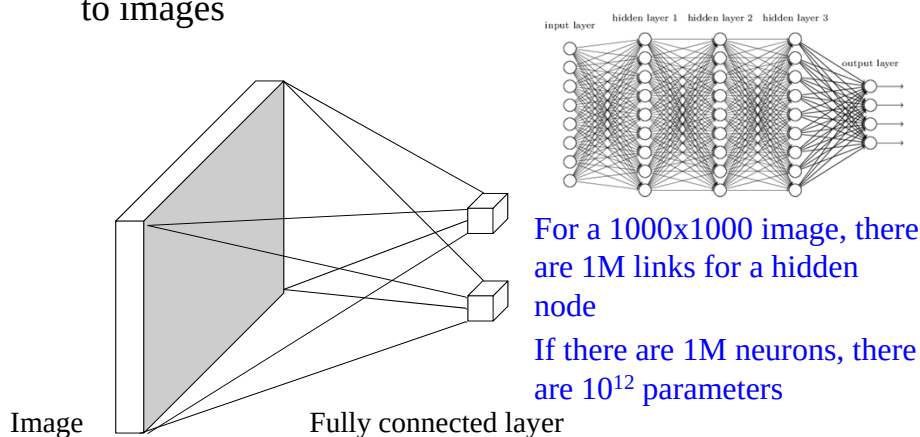


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## Deep Learning for Object Recognition

- Too many links when applying fully connected layers to images



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## Convolutional Neural Networks (CNN)

- CNN: a multi-layer neural network with
  1. **Local connectivity**
  2. **Weight sharing**
- Why local connectivity?
  - Spatial correlation is local (**locality of spatial dependencies**)
  - Reduce # of parameters
- Why weight sharing?
  - Statistics is at different locations (**stationarity of statistics**)
  - Reduce # of parameters

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## # of Parameters in Fully Connected NN

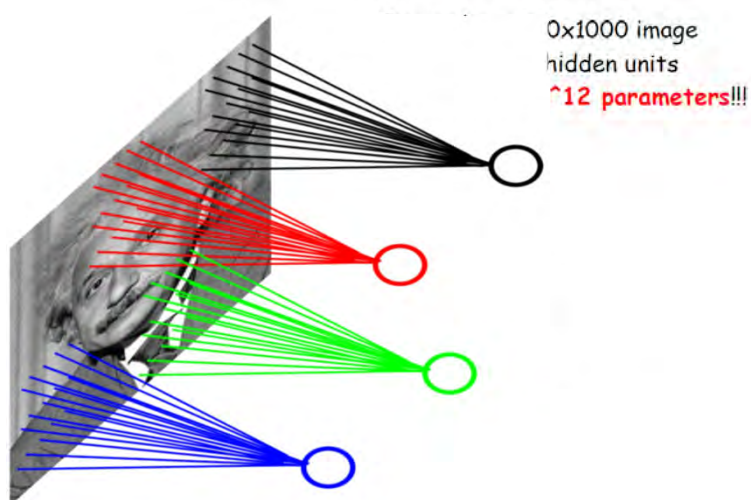
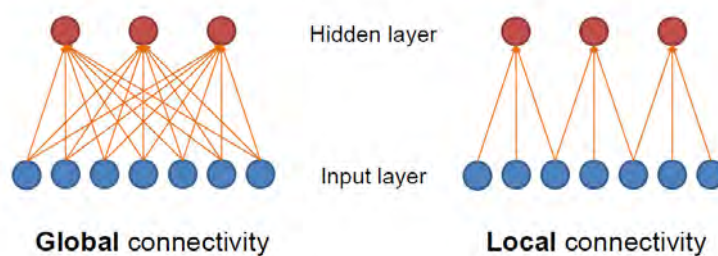


Image Credit: MA Ranzato

## CNN: Local Connectivity

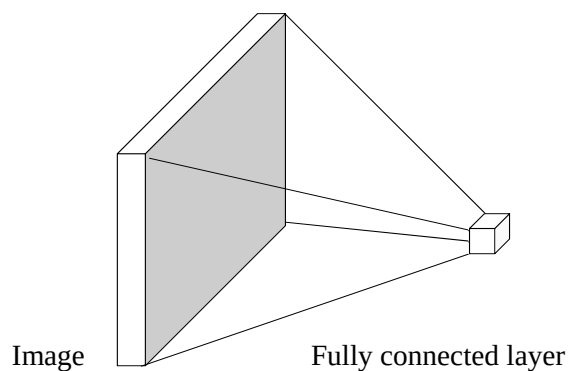


- # input units (neurons): 7
- # hidden units: 3
- Number of parameters
  - Global connectivity:  $3 \times 7 = 21$
  - Local connectivity:  $3 \times 3 = 9$

Image Credit: J.-B. Huang

## Local Connectivity

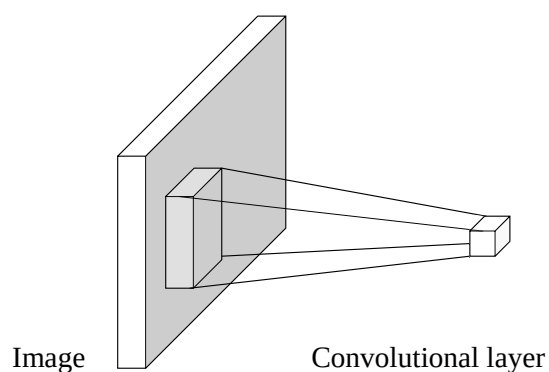
- Each neuron in the convolutional layer is connected to a local region, instead of the whole input image



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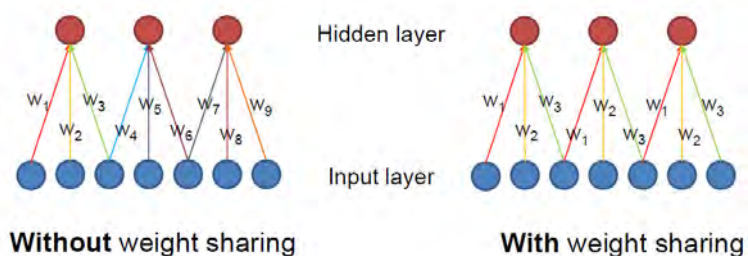
## Local Connectivity

- Each neuron in the convolutional layer is connected to a local region, instead of the whole input image



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## CNN: Weight Sharing



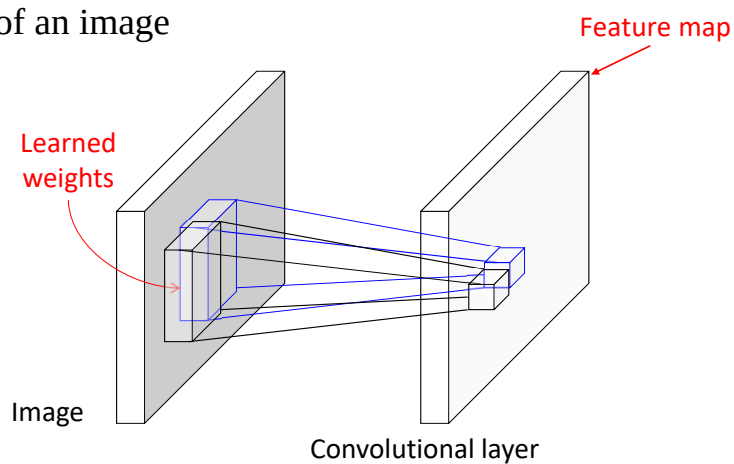
- # input units (neurons): 7
- # hidden units: 3
- Number of parameters
  - Without weight sharing:  $3 \times 7 = 21$
  - With weight sharing:  $3 \times 3 = 9$

Image Credit: J.-B. Huang

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## Weight Sharing

- We compute the same statistics at different locations of an image



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## Filtering

- How a filter works on an image?

|                        |                        |                        |   |   |
|------------------------|------------------------|------------------------|---|---|
| 1<br><small>x1</small> | 1<br><small>x0</small> | 1<br><small>x1</small> | 0 | 0 |
| 0<br><small>x0</small> | 1<br><small>x1</small> | 1<br><small>x0</small> | 1 | 0 |
| 0<br><small>x1</small> | 0<br><small>x0</small> | 1<br><small>x1</small> | 1 | 1 |
| 0                      | 0                      | 1                      | 1 | 0 |
| 0                      | 1                      | 1                      | 0 | 0 |

Image

|   |  |  |
|---|--|--|
| 4 |  |  |
|   |  |  |
|   |  |  |

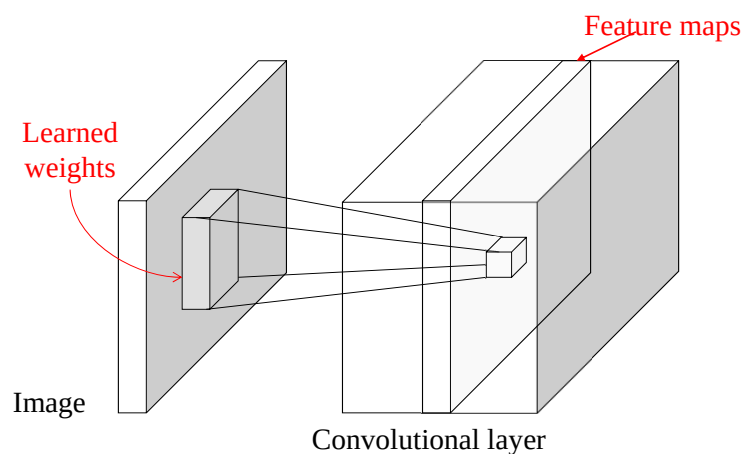
Convolved  
Feature

[http://ufldl.stanford.edu/wiki/images/6/6c/Convolution\\_schematic.gif](http://ufldl.stanford.edu/wiki/images/6/6c/Convolution_schematic.gif)

- Applying a filter to an image yields one feature map
- Applying  $N$  filters to an image yields  $N$  feature maps

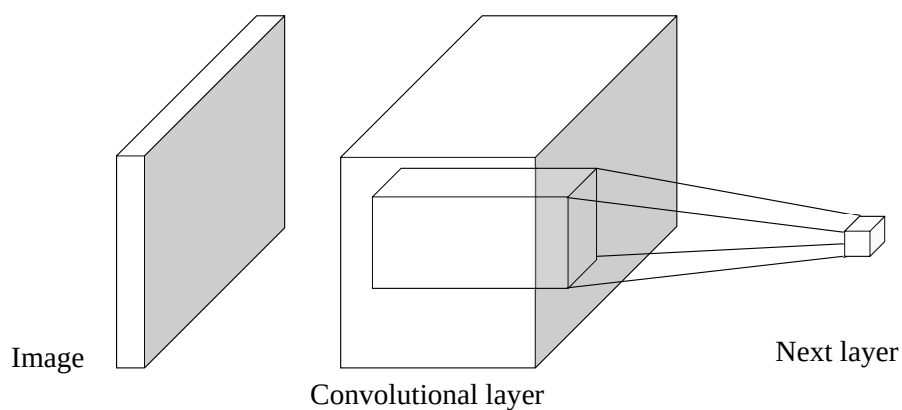
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## Multiple Feature Maps by Learning Multiple Filters



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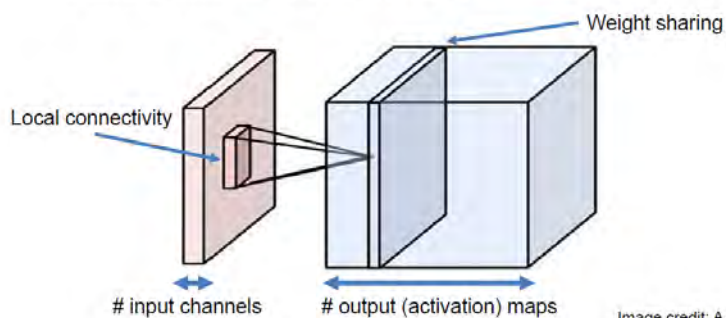
## Stacking Convolutional Layers



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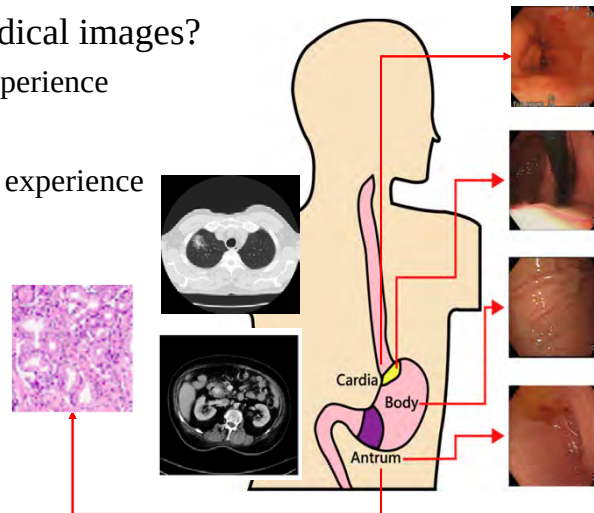
## Putting them together

- Local connectivity
- Weight sharing
- Handling multiple input channels
- Handling multiple output maps

Image credit: A. Karpathy  
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## Computational Medicine

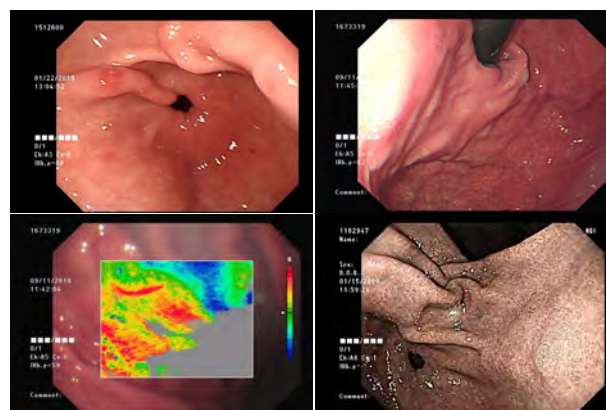
- Interpret medical images?
  - Experts' experience
- AI models?
  - Learn their experience



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## Gastric Section Classification

- Different gastric sections provide degrees and disease information



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## Gastric Section Classification

### Endoscopic Image Recording

During upper gastrointestinal (GI) endoscopy, endoscopic images of the stomach and duodenum are taken to record the severity of inflammation, malignant lesions, erosions and ulcers

### Diagnosis Evidence

These images serve as the diagnosis evidence of the patient and are used to describe the patient's gastric conditions

### Clinical Observation

By observing color and texture changes of blood vessels, and irregularities and deformities of mucosa from endoscopic images of these gastric sections, physicians can diagnose diseases

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## Gastric Section Classification

### Precancerous Markers

Corpus-predominant gastritis index (CGI) and spasmolytic polypeptide expressing metaplasia (SPeM) identify high gastric cancer risk patients from different gastric sections — the **antrum, body, and cardia**

### Growing Challenge

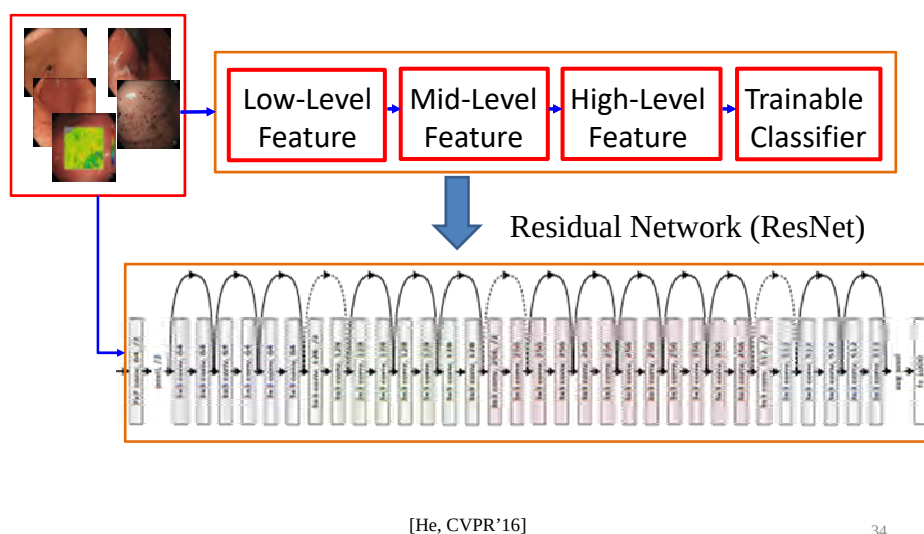
With the increasing number of endoscopic images, manual selection of endoscopic images for aforementioned medical applications is **time-consuming**

⚠ Manual review of endoscopic images is no longer scalable — AI automation is essential.

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## Usage of CNN



## Will Deep Learning Be Better?

- Yes

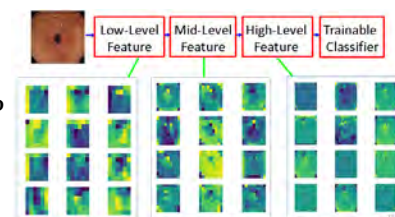
| Feature Type  | Method | Accuracy | Precision | Recall | F1-Score | Time (s) |
|---------------|--------|----------|-----------|--------|----------|----------|
| Conventional  | MLP    | 0.721    | 0.642     | 0.642  | 0.641    | 0.001    |
|               | SVM    | 0.752    | 0.686     | 0.693  | 0.687    | 0.026    |
| Deep Learning | VGG19  | 0.830    | 0.774     | 0.772  | 0.771    | 0.004    |
|               | ResNet | 0.842    | 0.804     | 0.819  | 0.810    | 0.006    |

- Moreover, fast to compute because of GPU
- Good enough?

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## Ensemble of CNNs

- Can we use more weak learners for endoscopic image classification?
  - Sure
- How about using stronger learners?
  - Sure
- Hypothesis:
  - Different CNN architectures learn different levels of semantic image representation
- An ensemble of CNNs will enable higher quality features to be extracted



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## Naïve Idea

- The output of three CNN models are fused by the decision network constructed by three fully connected layers

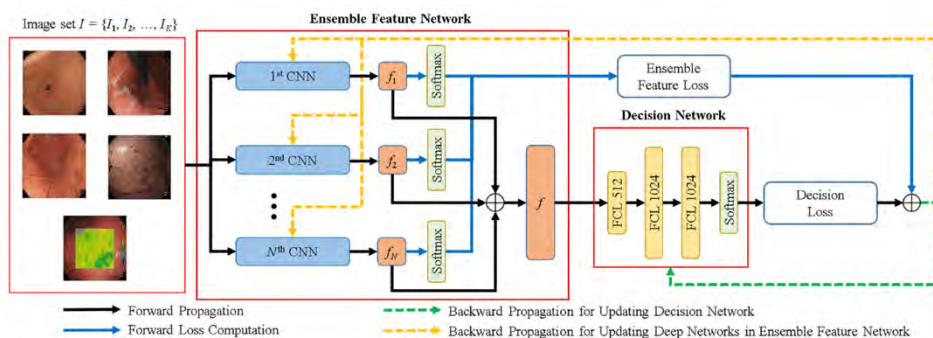


<https://ieeexplore.ieee.org/document/9107452>

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## Deep Ensemble Feature Network

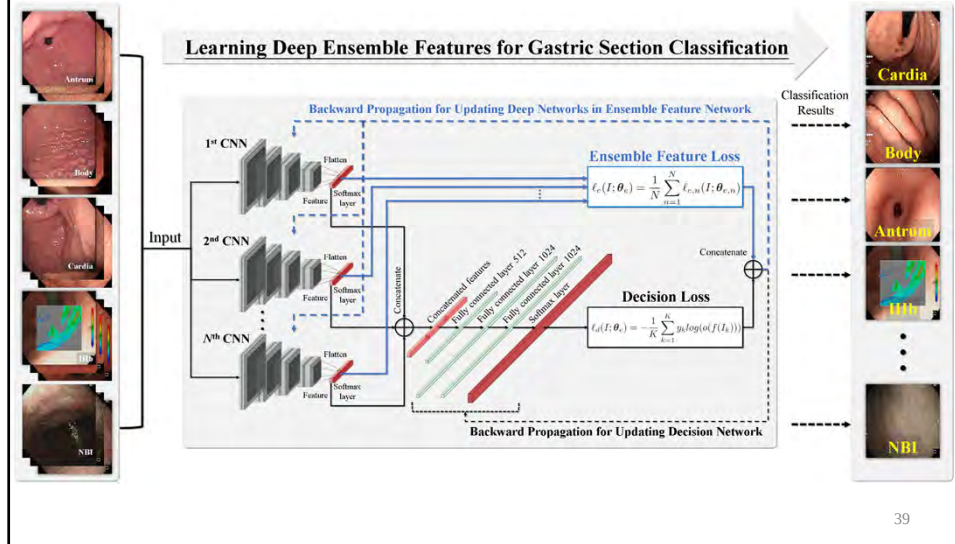
- Better idea?
  - Combining multiple CNN architectures in an end-to-end trainable framework for superior feature extraction and classification



<https://ieeexplore.ieee.org/document/9107452>

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# Deep Ensemble Feature Network



# Deep Ensemble Feature Network

## Two Sub-Networks

The DEF network is composed of two sub networks including the **ensemble feature network** and the **decision network**

## Ensemble Feature Network

Composed of  $N$  different CNNs with fully-connected layers and softmax layers, and generates the ensemble feature loss from CNNs for feature learning

## Decision Network

Consists of three fully connected layers followed by a softmax layer. For each image, the decision network estimates the label based on the deep ensemble feature

## Ensemble Feature Loss

### Purpose

The ensemble feature loss guides the training of the ensemble feature network by referring to classification labels and deep features computed by each CNN

### Design Basis

This loss is designed based on the **cross entropy** of all CNNs in the ensemble feature network

- The loss of the  $n$ th CNN can be defined as follows

$$\ell_{e,n}(I; \theta_{e,n}) = -\frac{1}{K} \sum_{k=1}^K y_k \log(o_n(f_n(I_k))), \quad (2)$$

where  $\theta_{e,n}$  is the parameter set of the  $n$ th CNN,  $o_n(f_n(I_k))$  is the softmax output of the  $n$ th deep feature of the  $n$ th CNN.

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## Ensemble Feature Loss

### Combined Ensemble Loss

With the losses of each CNN, the ensemble feature loss is defined as follows, where  $N$  is the number of CNNs in the ensemble feature network and is used to normalize the loss

### Eq. (2) and Eq. (3)

Learn the deep features of the individual CNNs and the concatenated deep ensemble feature from the training dataset  $I$ , respectively

$$\ell_e(I; \theta_e) = \frac{1}{N} \sum_{n=1}^N \ell_{e,n}(I; \theta_{e,n}), \quad (3)$$

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## Decision Loss

### Problem 1

The classification results decided by each CNN may be **inconsistent**

### Problem 2

The **fusion of different deep features** is not considered when using ensemble feature loss alone

### Solution: Decision Loss

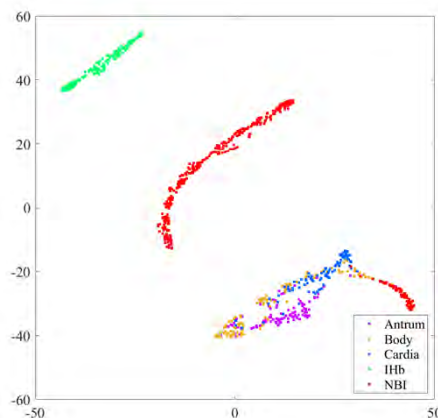
To learn the parameters of the decision network, the decision loss is defined to address both problems simultaneously

$$\ell_d(I; \theta_e) = -\frac{1}{K} \sum_{k=1}^K y_k \log(o(f(I_k))), \quad (4)$$

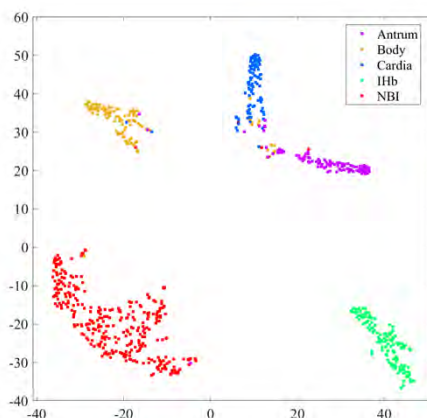
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## Visualization of Features

VGG

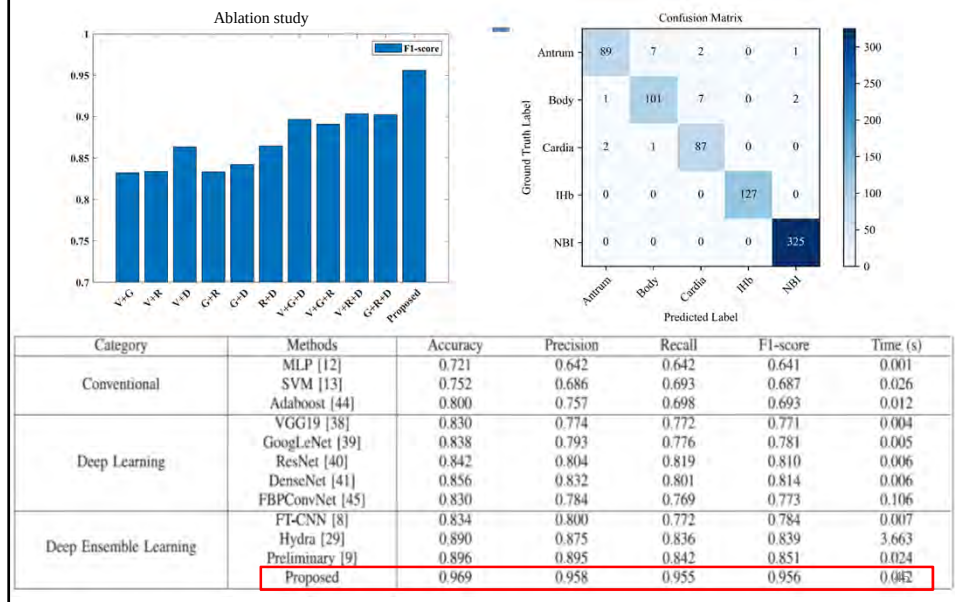


DEF Network



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## Will Deep Ensemble Learning Be Better?

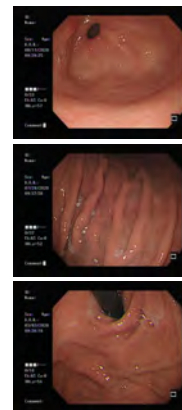


## Gastric Precancerous Lesion Diagnosis

- Corpus-predominant gastritis index (CGI)
  - The sum of the acute inflammation score (AIS) and chronic inflammation score (CIS) of antrum, body and cardia

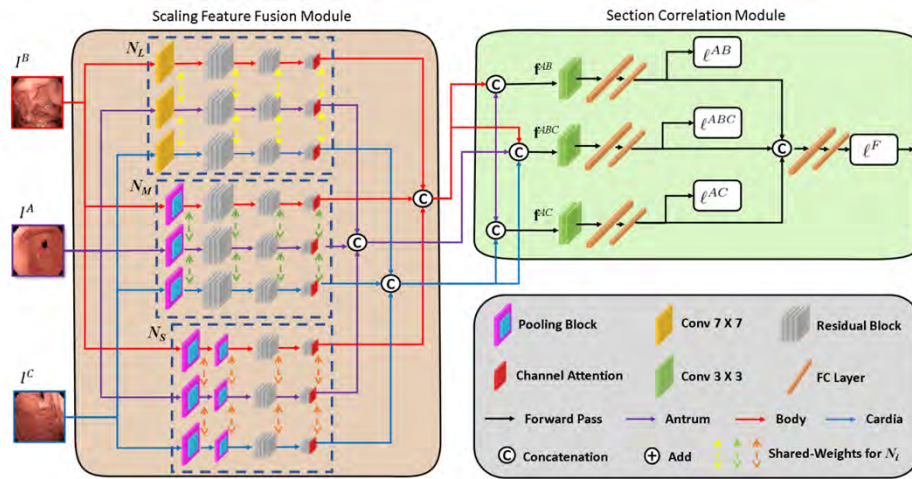
$$CGI = \begin{cases} 1, & \text{if } (IS_A < IS_B) \\ 1, & \text{if } (IS_A \leq IS_C \text{ and } IS_C \neq 1) \\ 0, & \text{otherwise} \end{cases}$$

- Current limitations?
  - Invasive biopsy
  - Time consuming pathology reports





## Gastric Section Correlation Network



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## Scaling Feature Fusion Module

### Multi-Scale Extraction

Simultaneously extract gastric section features with respect to **different scales and field-of-views** for different gastric sections

### Shared-Weights Sub-Networks

Contains three individual shared-weights resizing sub-networks, which can automatically achieve the resizing process by using the proposed network structure

### Variable Depth

Each subnetwork also contains **different numbers of convolutional layers and residual blocks** to capture features at varying levels of abstraction

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## Section Correlation Module

### Core Function

The section correlation module combines scaling fusion features of different gastric sections and section correlation losses for the **CGI diagnosis** from endoscopic images

### Heterogeneous Features

Combines scaling fusion features of different gastric sections to generate **heterogeneous section features** that capture cross-section relationships

### Shared-Weights Architecture

Contains three individual shared-weights resizing sub-networks which can automatically achieve the resizing process by using the proposed network structure

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## Quantitative Results

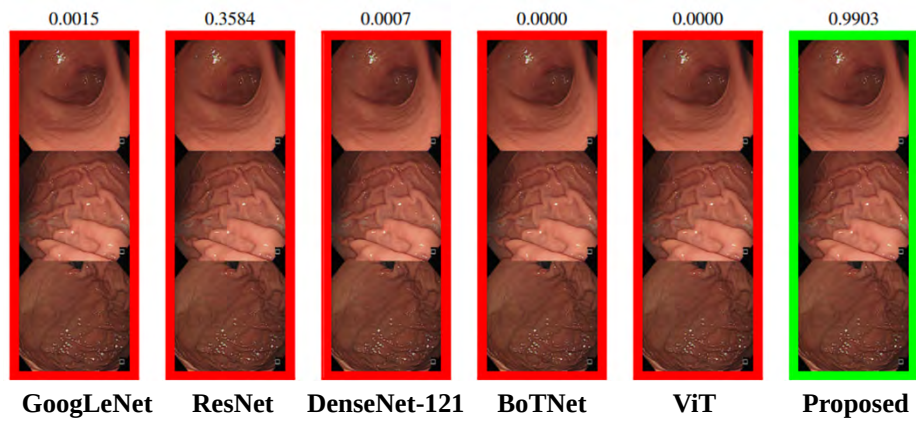
- 306 patients

| # of Patients | With CGI | Without CGI | Total |
|---------------|----------|-------------|-------|
| Training      | 63       | 149         | 212   |
| Testing       | 16       | 78          | 94    |

| Method       | Acc.          | Sens.         | Spec.         | AUC           |
|--------------|---------------|---------------|---------------|---------------|
| GoogLeNet    | 0.8723        | 0.6875        | 0.9103        | 0.8726        |
| ResNet-50    | 0.8936        | 0.6875        | 0.9487        | 0.9195        |
| DenseNet-121 | 0.8408        | 0.7500        | 0.8590        | 0.8766        |
| BoTNet       | 0.8617        | 0.8125        | 0.8718        | 0.8950        |
| ViT          | 0.7979        | 0.7500        | 0.8077        | 0.8910        |
| Proposed     | <b>0.9574</b> | <b>0.9375</b> | <b>0.9615</b> | <b>0.9836</b> |

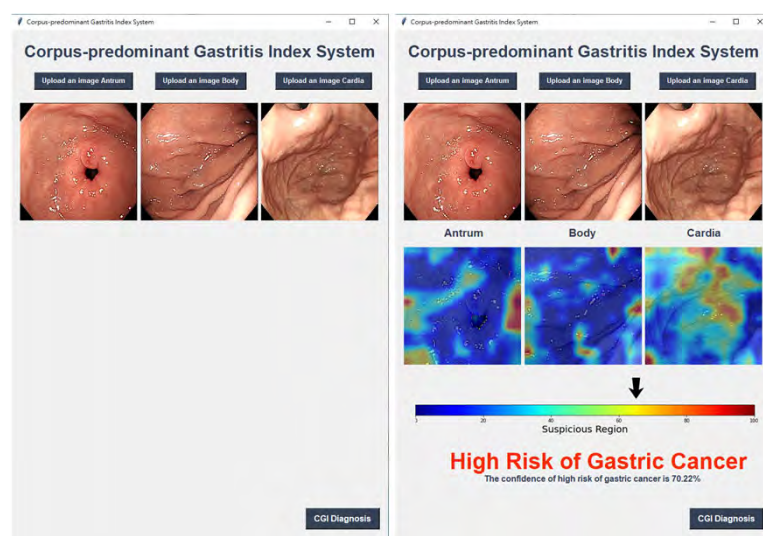
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## Qualitative Results



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## What Does AI Focus On?



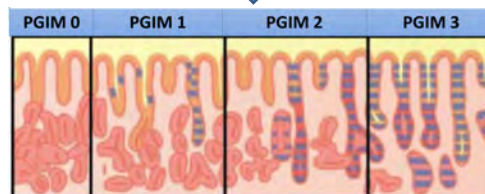
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## Gastric Precancerous Lesion Diagnosis

- Pathological gastric intestinal metaplasia (PGIM) grading by biopsies



↓ Microscopy (Five biopsies parts)

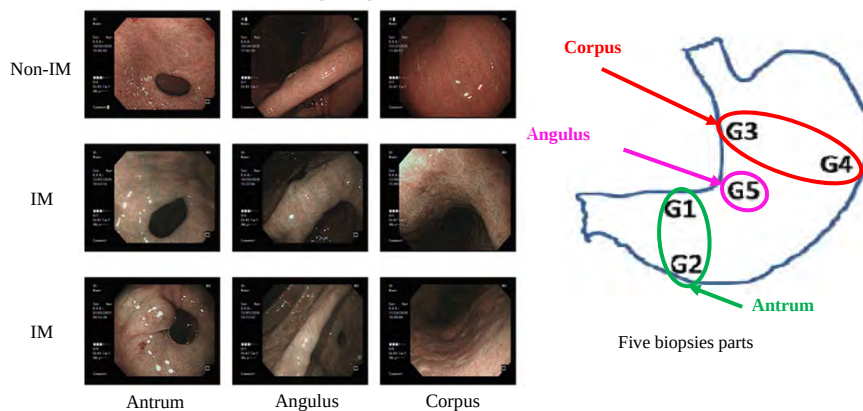


Ref.1: Project of AIGIM proposed by Wei-Lun Chang (national cheng kung university hospital)  
Ref.2: [https://colorectalsurgeonsydney.com.au/sites/default/files/inline-images/gastroscopy\\_u.jpg](https://colorectalsurgeonsydney.com.au/sites/default/files/inline-images/gastroscopy_u.jpg)

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## Gastric Precancerous Lesion Diagnosis

- Endoscopic grading of gastric IM (EGGIM) using narrow band imaging (NBI)



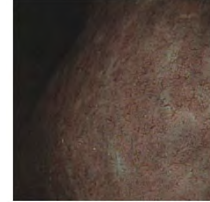
Ref.: Project of AIGIM proposed by Wei-Lun Chang (national cheng kung university hospital)

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# Issues

## Missed GIM

GIM could be missed because of **subtle mucosal changes** and **unclear boundaries**



## Camera Variability

The viewing directions and positions of the endoscopic camera introduce significant variability

## Mucosa Deformation

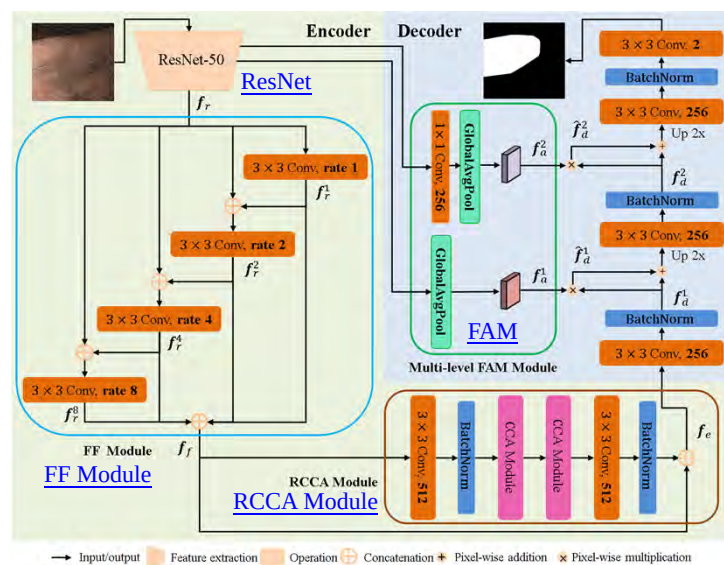
The mucosa deforms with time due to the **peristalsis of the stomach**

## Expert Dependency

Results rely heavily on **experienced doctors**, limiting scalability and consistency

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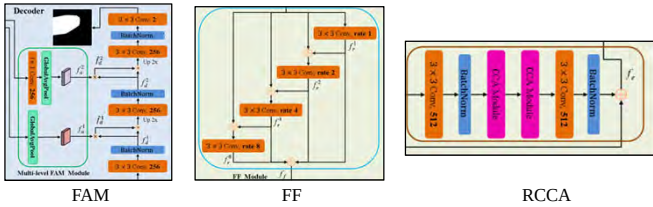
# Criss-Cross Attention based Multi-Level Fusion Network



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# Ablation Study

| FAM | FF          | RCCA | mIoU         | mDice        | Recall       | Precision    | Accuracy     |
|-----|-------------|------|--------------|--------------|--------------|--------------|--------------|
| ✓   | ✓           |      | 66.93        | 76.54        | 73.45        | 80.79        | 95.75        |
| ✓   |             | ✓    | 66.44        | 76.07        | 73.53        | 79.37        | 95.56        |
|     | ✓           | ✓    | 65.97        | 75.54        | 71.93        | 80.91        | 95.69        |
| ✓   | ✓ $\ominus$ | ✓    | 64.02        | 73.45        | 69.76        | 79.29        | 95.42        |
| ✓   | ✓           | ✓    | <b>68.92</b> | <b>78.47</b> | <b>74.94</b> | <b>83.45</b> | <b>96.13</b> |



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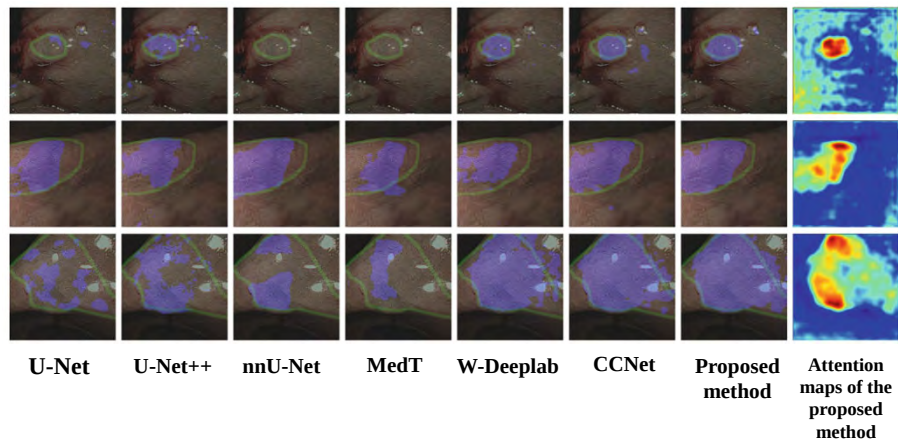
# Quantitative Results

– Comparisons with state-of-the-art methods (%)

| Method          | Backbone         | mIoU         | mDice        | Recall       | Precision    | Accuracy     |
|-----------------|------------------|--------------|--------------|--------------|--------------|--------------|
| U-Net           | U-Net            | 57.61        | 65.60        | 61.40        | 77.18        | 94.94        |
| U-Net++         | U-Net            | 58.62        | 66.92        | 62.55        | 78.10        | 95.05        |
| nnU-Net         | U-Net            | 60.33        | 69.05        | 64.29        | 80.47        | 95.30        |
| MedT            | Transformer      | 53.31        | 59.44        | 56.62        | 71.03        | 94.46        |
| W-Deeplab       | Deeplab-v3       | 62.19        | 71.22        | 65.66        | <b>85.17</b> | 95.69        |
| CCNet           | Resnet-50        | 65.88        | 75.39        | 70.80        | 83.24        | 95.87        |
| <b>Proposed</b> | <b>Resnet-50</b> | <b>68.92</b> | <b>78.47</b> | <b>74.94</b> | 83.45        | <b>96.13</b> |

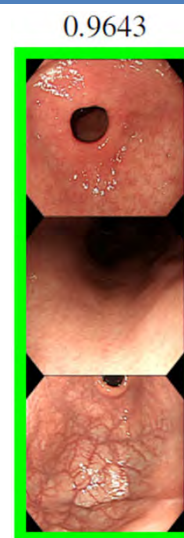
58

## Qualitative Results



## Summary

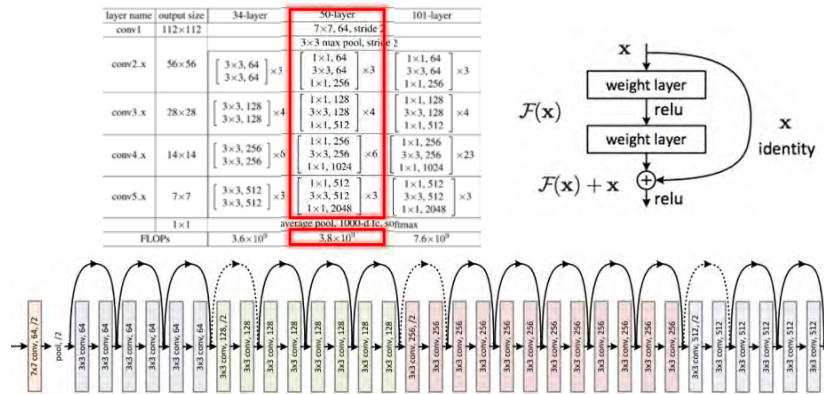
- AI for endoscopic image analysis
  - Conventional techniques
    - Learn experts' experience from hand-crafted features
  - Recent deep learning techniques
    - Learn experts' experience from images
    - Generate representative features
- More applications
  - In colonoscopy
  - In bronchoscopy





## Criss-Cross Attention based Multi-Level Fusion Network

- Backbone deep neural network-ResNet50

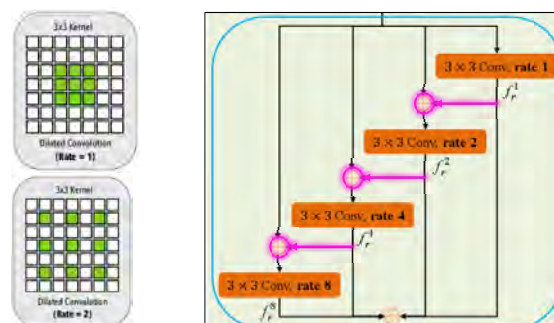


Ref.: [https://jananisabhu.github.io/ResNet50\\_From\\_Scratch\\_Tensorflow/](https://jananisabhu.github.io/ResNet50_From_Scratch_Tensorflow/)

61

## Criss-Cross Attention based Multi-Level Fusion Network

- Feature fusion (FF) module
  - Dilated convolution
  - Concatenation with different dilated rates

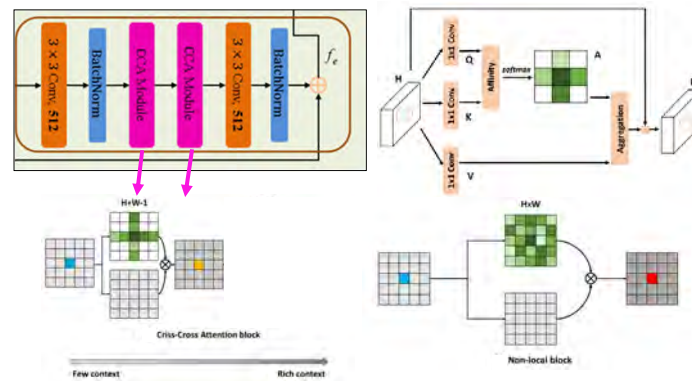


Ref.: [https://www.researchgate.net/figure/Dilated-convolution-On-the-left-we-have-the-dilated-convolution-with-dilation-rate-r\\_fig2\\_320195101](https://www.researchgate.net/figure/Dilated-convolution-On-the-left-we-have-the-dilated-convolution-with-dilation-rate-r_fig2_320195101)

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## Criss-Cross Attention based Multi-Level Fusion Network

- Recurrent criss-cross attention (RCCA) module

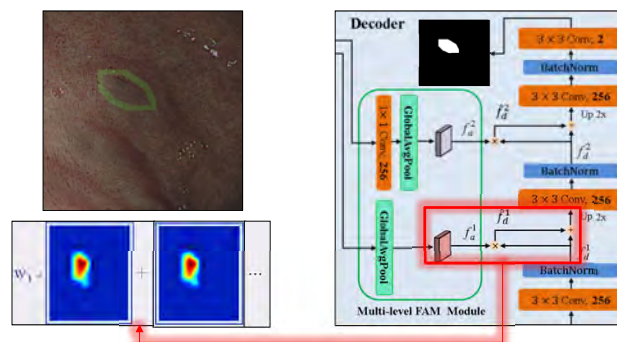


Ref.: <https://arxiv.org/abs/1811.11721>

63

## Criss-Cross Attention based Multi-Level Fusion Network

- Multi-level feature activation map (FAM) module



Ref.: <https://arxiv.org/abs/1512.04150>

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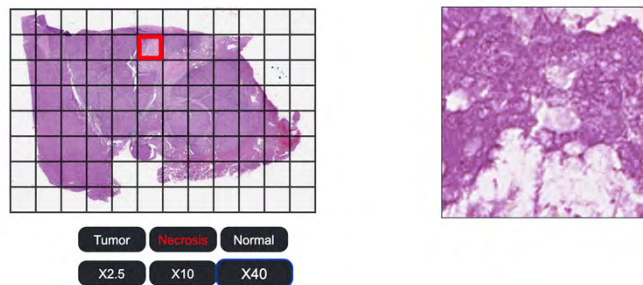
## Pathology Image Analysis

### Annotation Challenge

To achieve AI methods for digital pathology analysis, one of the most challenging problems is the **difficulty of obtaining accurate and sufficient annotation data**

### Manual Labeling Burden

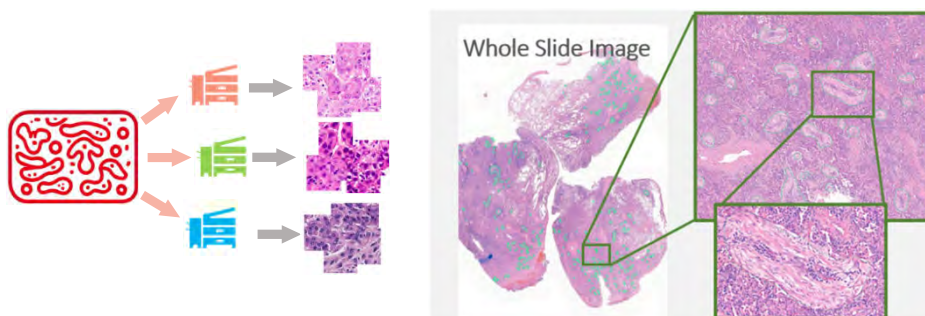
These annotations require **manually time-consuming labeling process** by the pathologists, creating a significant bottleneck for AI development



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## Pathology Image Analysis

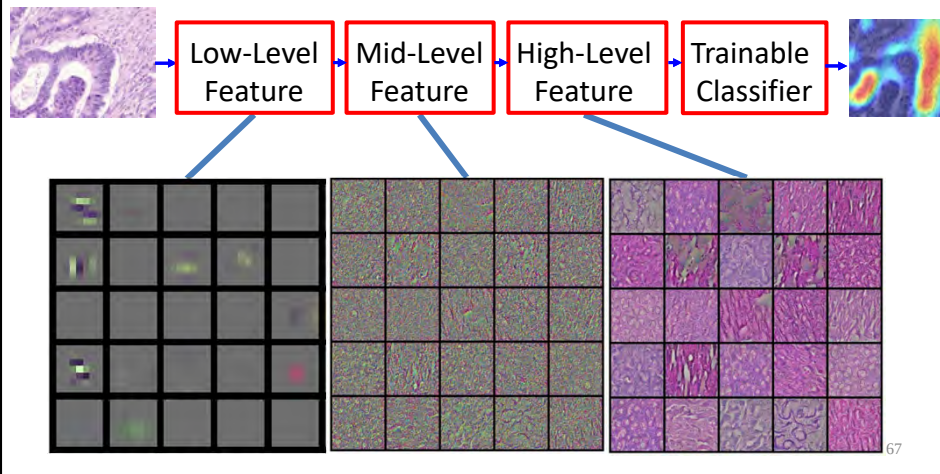
- Tissues from different organs
- Extremely high resolutions of whole slide images
- Variant scanning magnification factors



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## Deep Learning for Pathology Images ?

- Also learn hierarchical representations



## Insufficient Training Data

### Partial Annotation Problem

Pathologists may only be able to label a portion of images and leave many **unlabeled regions**, creating incomplete training

### Key Research Question

How to use **non-annotated data** to help model training becomes an important issue in computational pathology

### Semi-Supervised Learning

Both annotated and unannotated data are used for training

### Self-Supervised Learning

A form of unsupervised learning where the data themselves provide the supervision

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## Semi-supervised Learning

- Semi-supervised learning methods generate pseudo-labels of unlabeled data and utilize the information from both labeled and unlabeled data to reduce the required amount of labeled data for training

### Pseudo Labels

Using a well-trained model to generate **non-manually-crafted labels** (pseudo labels) for unlabeled data, expanding the effective training set

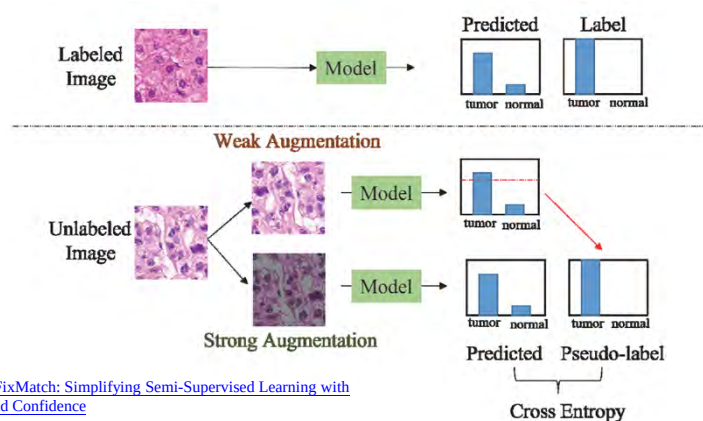
### Consistency Regularization

Training a model that is **invariant to various data augmentations**, enforcing consistent predictions across different views of the same image

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## Semi-supervised Learning

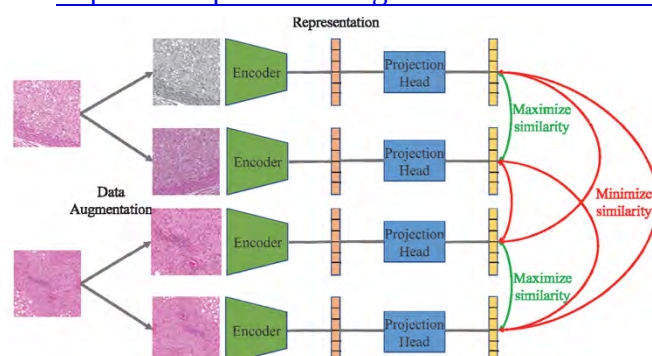
- Classical method combining consistency regularizations and pseudo labels is FixMatch



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## Self-supervised Learning

- Contrastive learning
  - Learning the representation of a dataset
- The milestone of contrastive learning is SimCLR
  - AI-X <https://ieeexplore.ieee.org/document/10384535>



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## Computational Complexity for Whole Slide Imaging

- The resolution of the pathology whole slide image (WSI) is significantly higher than those of the natural images

### Patch-wise Classification

Reduce computation by using **patch-wise classification** instead of semantic segmentation — processing manageable image tiles

### Unique Network Designs

Develop unique strategy or network designs for the characteristics of symptoms such as **multiscale feature extraction**

### System Structure Modification

Modify system structure to **increase inference speed** in cell level segmentation for real-time clinical applications

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# Pathology Image Classification

## Histopathological Image Classification

Histopathological images provide the **medical evidences** to help the disease diagnosis across a wide range of conditions

## Time-Consuming Manual Review

However, manually reviewing these images by pathologists is very **time-consuming**, creating a bottleneck in clinical workflows

## Classification Challenges

- Different organs with distinct tissue morphologies
- Different magnification factors across scanning protocols

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# Evaluations of Deep Learning Methods

- Deep learning methods are designed for natural image classification



### VGG Network

Deep convolutional network with uniform architecture



### Residual Neural Network

Skip connections enabling very deep networks



### DenseNet

Densely connected convolutional network



### DEFNet

Deep Ensemble Feature Network



### Vision Transformer

Self-attention based image recognition



### Swin Transformer

Shifted window transformer architecture



### Bottleneck Transformer

Hybrid CNN-Transformer with bottleneck attention

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## DenseNet

- Dense block: a few densely connected convolutional layers
- Stack dense blocks
- Insert a convolutional layer and a pooling layer into two connected blocks

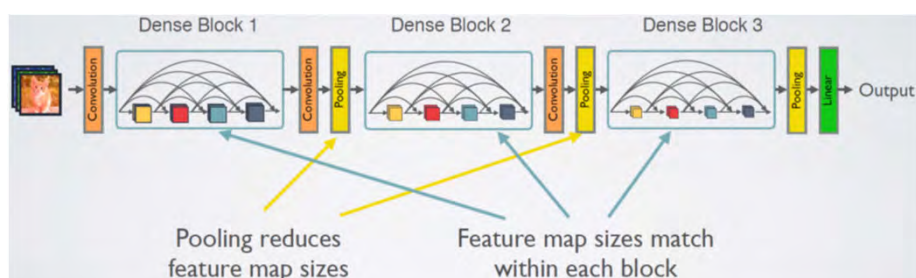
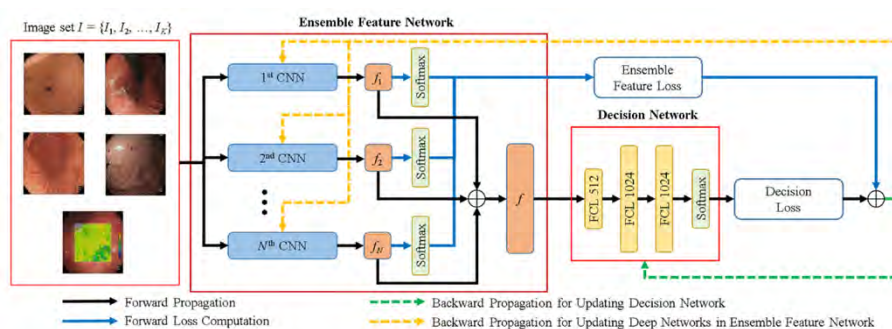


Photo: Huang et al.

75

## Deep Ensemble Feature Network

- Ensemble different CNN networks



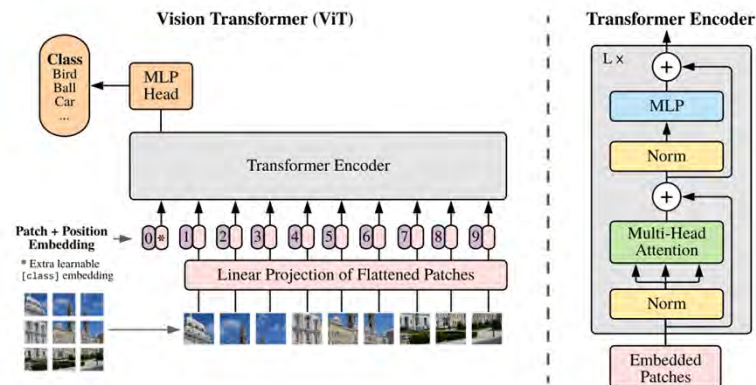
<https://ieeexplore.ieee.org/document/9107452>

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# Vision Transformer Architecture

- Self-attention structure



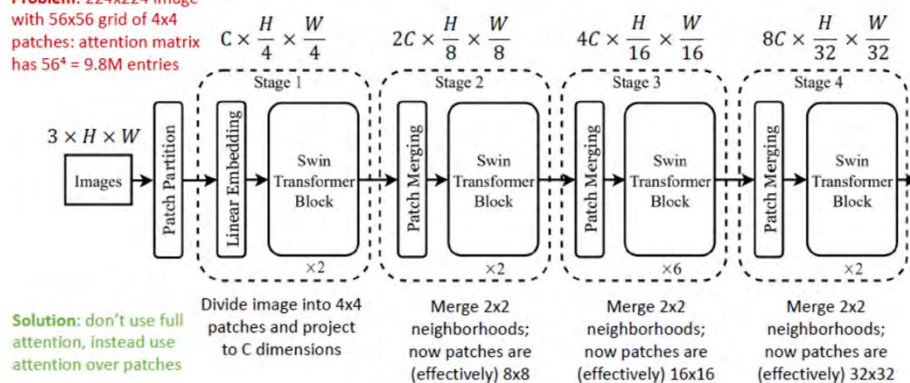
<https://arxiv.org/abs/2010.11929>

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# Swin Transformer

- Self-attention with shift windows

**Problem:** 224x224 image  
with 56x56 grid of 4x4  
patches: attention matrix  
has  $56^4 = 9.8M$  entries

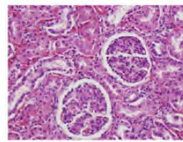


[2103.14030] Swin Transformer: Hierarchical Vision Transformer using Shifted Windows

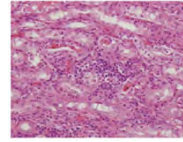
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## Result of ADL Dataset

|              | Kidney       | Lung         | Spleen       |
|--------------|--------------|--------------|--------------|
| Methods      | Acc. (%)     | Acc. (%)     | Acc. (%)     |
| VGGNet19 [1] | 90.98        | 94.30        | 89.16        |
| ResNet [2]   | 94.51        | 96.05        | 94.58        |
| DenseNet [3] | 96.47        | 96.92        | 92.50        |
| DEFNet [4]   | 97.66        | 97.50        | 97.92        |
| ViT [5]      | 97.25        | 94.30        | 95.42        |
| SwinT [6]    | <b>98.43</b> | <b>98.25</b> | <b>98.33</b> |
| BoTNet [7]   | 92.16        | 98.25        | 94.16        |



Healthy kidney

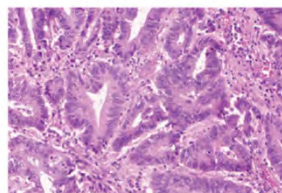


Inflamed kidney

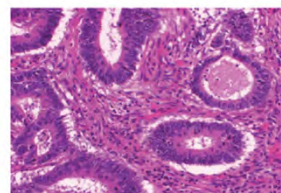
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## Result of GLAS Dataset

| Methods      | Acc. (%) |
|--------------|----------|
| VGG [1]      | 87.50    |
| ResNet [2]   | 91.25    |
| DenseNet [3] | 88.75    |
| DEFNet [4]   | 95.00    |
| ViT [5]      | 92.50    |
| SwinT [6]    | 97.50    |
| BoTNet [7]   | 78.75    |



Benign



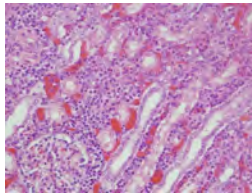
Malignant

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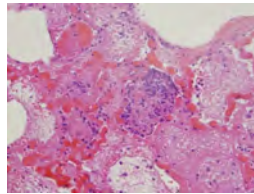


## More Issues

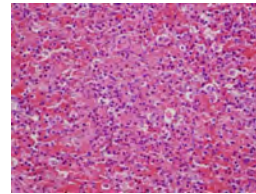
- Sample histopathological images
  - Different organs



Kidney



Lung



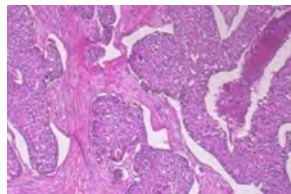
Spleen

U. Srinivas, H. S. Mousavi, V. Monga, A. Hattel, and B. Jayarao, "Simultaneous sparsity model for histopathological image representation and classification," *IEEE Trans. on Medical Imaging*, 2014.

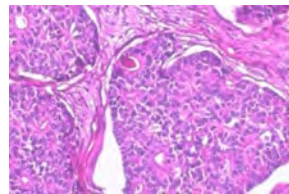
## More Issues

- Sample histopathological images
  - Different magnification factors

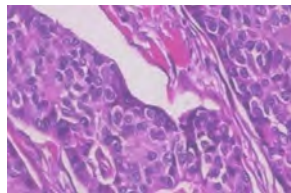
40×



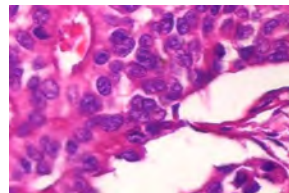
100×



200×



400×



82

F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "A dataset for breast cancer histopathological image classification," *IEEE Trans. on Biomedical Engineering*, 2016.

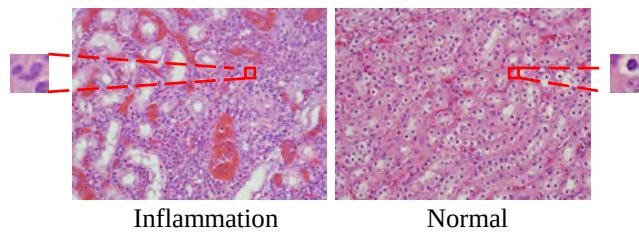
## More Issues

### 1. Hand-Crafted Features

Hard to effectively represent the **appearance variations** across different tissue types, organs, and magnification levels in histopathological images

### 2. CNN-Based Methods

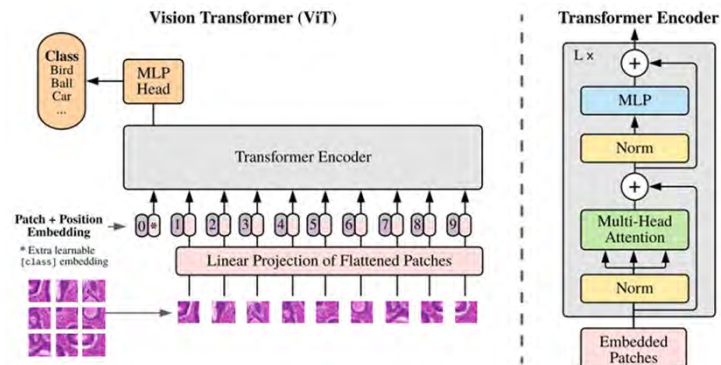
Field-of-view usually **limited by the kernel sizes of filters** — only the features of local regions are extracted, missing long-range tissue context



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## Thinking

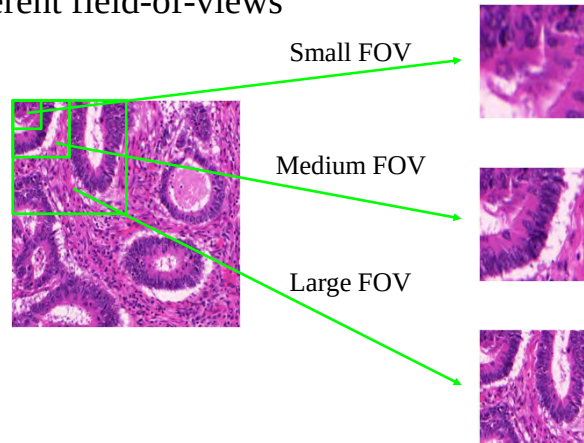
- Vision transformer
  - ViT is still under a fixed field-of-view with respect to the patch sizes



A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, <sup>84</sup>An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int'l Conf. Learning Representations*, 2021.

## More About Magnification Factors

- Different field-of-views

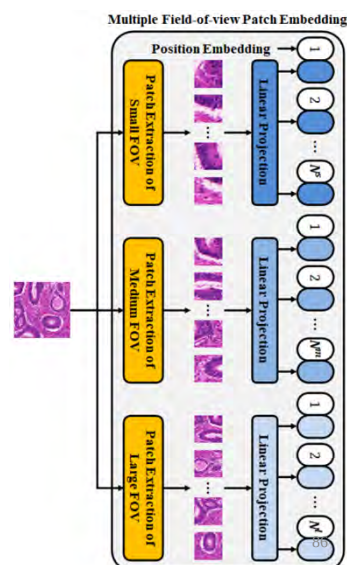


85

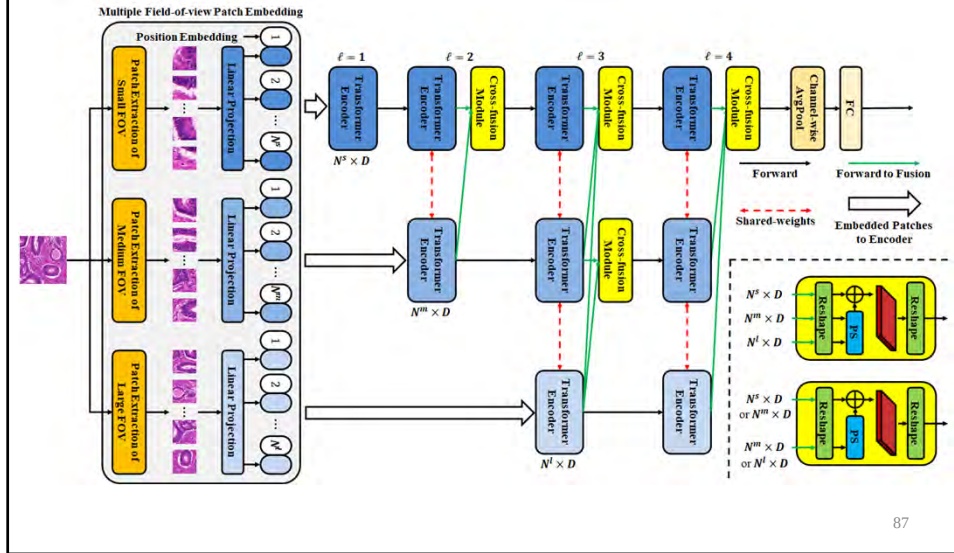
## More About Magnification Factors

- Represent patches from different field-of-views

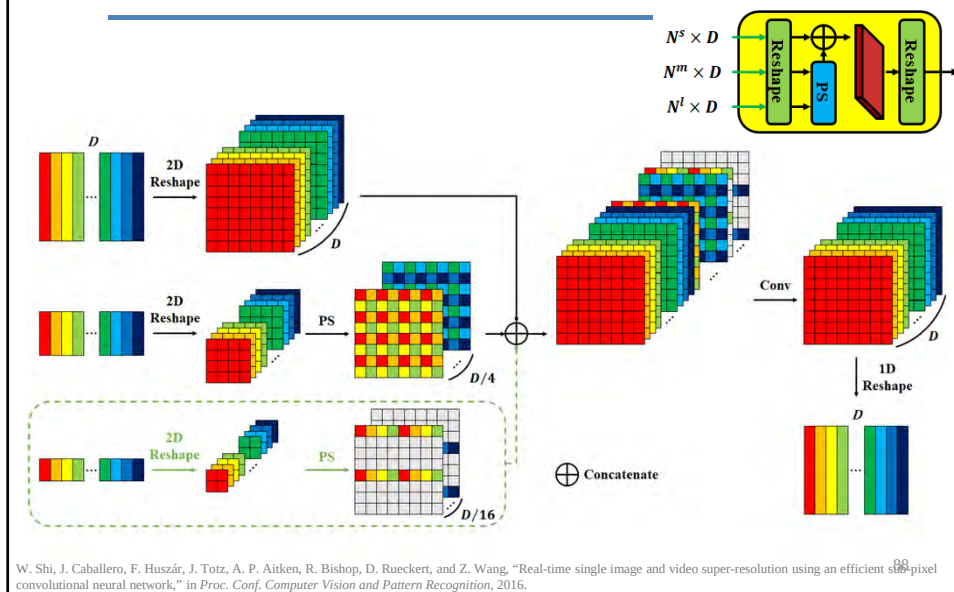
☞ Patches from Small FOV, Medium FOV, and Large FOV are each extracted and projected via linear projection with position embedding, enabling multi-scale representation.



## Cross-scale Fusion Transformer



## Cross-fusion Module



## BreakHis Dataset

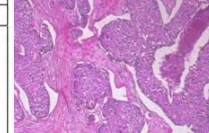
- Breast cancer

TABLE I

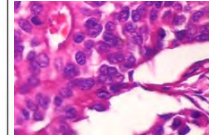
COMPARISONS OF THE BREAKHIS DATASET BY USING RECOGNITION RATES (%).

| Method              | Image Level  |              |              |              | Patient Level |              |              |              |
|---------------------|--------------|--------------|--------------|--------------|---------------|--------------|--------------|--------------|
|                     | 40×          | 100×         | 200×         | 400×         | 40×           | 100×         | 200×         | 400×         |
| Spanhol et al. [43] | -            | -            | -            | -            | 81.60         | 79.90        | 85.10        | 82.30        |
| Spanhol et al. [21] | 85.60        | 83.50        | 82.70        | 80.70        | 90.00         | 88.40        | 84.60        | 86.10        |
| FVCNN [22]          | 87.00        | 86.20        | 85.20        | 82.90        | 90.00         | 88.90        | 86.90        | 86.30        |
| Han et al. [24]     | 95.80        | 96.90        | 96.70        | 94.90        | 97.10         | 95.70        | 96.50        | 95.70        |
| Hou [12]            | 90.89        | 90.99        | 91.00        | 90.97        | 91.00         | 91.00        | 91.00        | 91.00        |
| Toğaçar et al. [25] | 97.99        | 97.84        | 98.51        | 95.88        | -             | -            | -            | -            |
| Li et al. [23]      | 96.00        | 96.16        | 98.01        | 95.97        | 95.01         | 96.84        | 97.92        | 96.28        |
| Mi et al. [26]      | 96.70        | 97.60        | 95.00        | 93.30        | -             | -            | -            | -            |
| Zou et al. [27]     | 97.58        | 97.47        | 99.08        | 96.52        | 96.79         | 97.41        | 99.29        | 97.16        |
| Burçak et al. [33]  | 97.00        | 97.00        | 96.00        | 96.00        | -             | -            | -            | -            |
| AlexNet-BC [29]     | 98.15        | 97.71        | 97.96        | 98.48        | -             | -            | -            | -            |
| ViT [8]             | 97.94        | 96.86        | 98.71        | 96.82        | 97.69         | 96.77        | 99.15        | 95.63        |
| Swin [39]           | 97.42        | 98.68        | 98.71        | 98.59        | 98.18         | 98.69        | 98.21        | 98.68        |
| DCET-Net [9]        | <b>99.00</b> | 98.08        | 99.34        | 98.72        | 98.88         | 97.94        | 99.23        | 99.03        |
| HIPT [37]           | 80.41        | 84.13        | 87.74        | 85.87        | 84.09         | 84.27        | 87.12        | 85.98        |
| Proposed            | 98.11        | <b>99.01</b> | <b>99.52</b> | <b>99.47</b> | <b>98.89</b>  | <b>99.22</b> | <b>99.58</b> | <b>99.46</b> |

40x



400x



89

F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "A dataset for breast cancer histopathological image classification," *IEEE Trans. on Biomedical Engineering*, 2016.

## YTMF Dataset

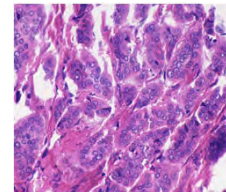
- Breast cancer

TABLE III

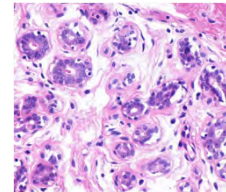
COMPARISONS OF THE YTMF DATASET BY USING THE CONFUSION MATRIX IN (%).

| Method         | Class     | Malignant    | Benign       |
|----------------|-----------|--------------|--------------|
| WND-CHARM [14] | Malignant | 46.15        | 53.85        |
| DFDL [2]       |           | 61.54        | 38.46        |
| SDL [18]       |           | 61.54        | 38.46        |
| ALSF [20]      |           | 65.38        | 34.62        |
| FVCNN [22]     |           | 65.38        | 34.62        |
| ViT [8]        |           | 61.54        | 38.46        |
| Swin [39]      |           | 65.38        | 34.62        |
| HIPT [37]      |           | 69.23        | 30.77        |
| Proposed       |           | <b>80.77</b> | <b>19.23</b> |
| WND-CHARM [14] | Benign    | 21.88        | 78.12        |
| DFDL [2]       |           | 15.62        | 84.38        |
| SDL [18]       |           | 9.38         | 90.62        |
| ALSF [20]      |           | 9.38         | 90.62        |
| FVCNN [22]     |           | 12.50        | 87.50        |
| ViT [8]        |           | 9.38         | 90.62        |
| Swin [39]      |           | 9.38         | 90.62        |
| HIPT [37]      |           | 18.75        | 81.25        |
| Proposed       |           | <b>6.25</b>  | <b>93.75</b> |

Malignant



Benign

L. E. Boucheron, B. S. Manjunath, and N. R. Harvey, "Use of imperfectly segmented nuclei in the classification of histopathology images of breast cancer," in *Proc. 9th Int. Conf. on Acoustics, Speech and Signal Processing*, 2010.

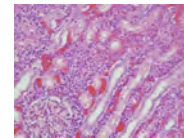


## ADL Dataset

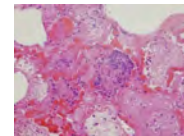
- Kidney, lung and spleen of bovine

TABLE II  
COMPARISONS OF THE ADL DATASET BY USING THE CONFUSION MATRIX IN (%).

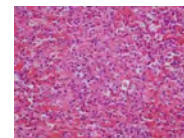
| Method                | Class        | Kidney       |             | Lung         |              | Spleen       |             |
|-----------------------|--------------|--------------|-------------|--------------|--------------|--------------|-------------|
|                       |              | Inflammation | Normal      | Inflammation | Normal       | Inflammation | Normal      |
| WND-CHARM [14]        | Inflammation | -            | -           | -            | -            | 88.2         | 11.8        |
| SHRC [11]             |              | 83.3         | 16.7        | 85.0         | 15.0         | 88.3         | 11.7        |
| SCD* [47]             |              | 82.3         | 17.7        | 41.0         | 59.0         | 65.0         | 35.0        |
| SLIDE* [48]           |              | 24.5         | 75.5        | 92.3         | 7.7          | 57.5         | 42.5        |
| DFDL [2]              |              | 90.0         | 10.0        | 97.4         | 2.6          | 92.0         | 8.0         |
| SDL* [18]             |              | 97.8         | 2.2         | 97.4         | 2.6          | 95.0         | 5.0         |
| ALSF [20]             |              | -            | -           | -            | -            | 96.6         | 3.4         |
| SDPSD + MIMCISM* [19] |              | 90.3         | 9.7         | 98.9         | 1.1          | 93.2*        | 6.2*        |
| FVCNN [22]            |              | -            | -           | -            | -            | 94.1         | 6.9         |
| DEFNet [49]           |              | 98.6         | 1.4         | 96.5         | 3.5          | 98.3         | 1.7         |
| ViT [8]               |              | 92.0         | 8.0         | <b>100.0</b> | <b>0.0</b>   | 97.5         | 2.5         |
| Swin [39]             |              | 98.6         | 1.4         | 97.4         | 2.6          | 98.3         | 1.7         |
| HIPT [37]             |              | 77.5         | 22.5        | 92.2         | 7.8          | 98.3         | 1.7         |
| Proposed              |              | <b>99.3</b>  | <b>0.7</b>  | <b>100.0</b> | <b>0.0</b>   | <b>99.2</b>  | <b>0.8</b>  |
| WND-CHARM [14]        | Normal       | -            | -           | -            | -            | 19.8         | 80.2        |
| SHRC [11]             |              | 17.5         | 82.5        | 25.0         | 75.0         | 35.0         | 65.0        |
| SCD* [47]             |              | 73.0         | 27.0        | 35.9         | 64.1         | 60.8         | 39.2        |
| SLIDE* [48]           |              | 15.0         | 85.0        | 67.3         | 32.7         | 56.1         | 43.9        |
| DFDL [2]              |              | 11.8         | 88.2        | 3.5          | 96.5         | 7.1          | 92.9        |
| SDL* [18]             |              | 12.5         | 87.5        | 2.6          | 97.4         | 9.8          | 90.2        |
| ALSF [20]             |              | -            | -           | -            | -            | 6.6          | 93.4        |
| SDPSD + MIMCISM* [19] |              | 9.9          | 90.1        | 1.0          | 99.0         | 5.8          | 94.2        |
| FVCNN [22]            |              | -            | -           | -            | -            | 10.9         | 90.1        |
| DEFNet [49]           |              | 3.4          | 96.6        | 1.8          | 98.2         | 2.5          | 97.5        |
| ViT [8]               |              | 9.4          | 90.6        | 11.5         | 88.5         | 6.6          | 93.4        |
| Swin [39]             |              | 1.7          | <b>98.3</b> | 0.9          | 99.1         | <b>1.6</b>   | <b>98.4</b> |
| HIPT [37]             |              | 11.1         | 88.9        | 16.8         | 83.2         | 12.4         | 87.6        |
| Proposed              |              | 3.4          | 96.6        | <b>0.0</b>   | <b>100.0</b> | <b>1.6</b>   | <b>98.4</b> |



Kidney



Lung



Spleen

U. Srinivas, H. S. Mousavi, V. Monga, A. Hattel, and B. Jayarao, "Simultaneous sparsity model for histopathological image representation and classification," *IEEE Trans. on Medical Imaging*, 2014.

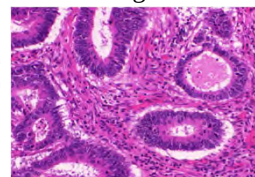
## GlaS Dataset

- Colorectal cancer

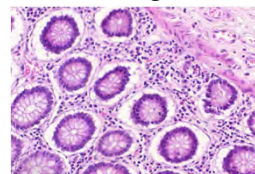
TABLE IV  
COMPARISONS OF THE GLAS DATASET BY USING ACCURACY (%).

| Method        | Accuracy      |
|---------------|---------------|
| ERASE [50]    | 92.50         |
| Wildcat [51]  | 98.75         |
| Deep MIL [52] | 97.50         |
| DEFNet [49]   | 95.00         |
| EEM [3]       | <b>100.00</b> |
| ViT [8]       | 92.50         |
| Swin [39]     | 97.50         |
| HIPT [37]     | 67.50         |
| Proposed      | <b>100.00</b> |

Malignant



Benign



K. Sirinukunwattana, D. R. J. Snead, and N. M. Rajpoot, "A stochastic polygons model for glandular structures in colon histology images," *IEEE Trans. on Medical Imaging*, 2015.

K. Sirinukunwattana, J. P. Pluim, H. Chen, X. Qi, P.-A. Heng, Y. B. Guo, L. Y. Wang, B. J. Matuszewski, E. Bruni, U. Sanchez *et al.*, "Gland segmentation in colon histology images: The glas challenge contest," *Medical image analysis*, 2017.

## Efficiency Comparisons

- Number of parameters and FLOPs

**TABLE V**  
PARAMETERS AND FLOPs

| Method    | Image size       | Params     | FLOPs        |
|-----------|------------------|------------|--------------|
| ViT [8]   | $224 \times 224$ | 86M        | 17.5G        |
| ViT [8]   | $384 \times 384$ | 86M        | 55.4G        |
| Swin [39] | $224 \times 224$ | 88M        | 15.4G        |
| Swin [39] | $384 \times 384$ | 88M        | 47.0G        |
| Proposed  | $256 \times 256$ | <b>57M</b> | <b>14.7G</b> |

### Parameters

Model size measured by total trainable parameters — a key indicator of memory requirements.

### FLOPs

Floating point operations per second — measures computational cost during inference.

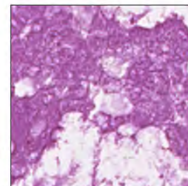
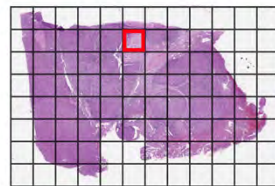
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## Pathology Image Segmentation

- Common liver cancer
  - Accounts for around 75%-85% of all liver cancer deaths
  - The WSIs of HCC liver tissues usually contain normal, tumorous, and necrotic regions



Try clicking the black buttons under the WSI to observe the different characteristics of the three kinds of regions at different magnifications.



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## Immersive Article

- <https://ieeexplore.ieee.org/document/10188451>

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ADVANCED SEARCH

Journals & Magazines > IEEE Computational Intelligence > Volume: 18 Issue: 3

### Multi-Magnification Attention Convolutional Neural Networks [AI-eXplained]

Publisher: IEEE Cite This PDF Immersive Article

Chia-Wei Chao ; Daniel Winden Hwang ; Hung-Wen Tsa ; Shih-Hsuan Lin ; Wei-Li Chen ; Chun-Rong Huang ; Pau-Choo Ch ; All Authors

6489 Full Text Views

**Abstract**

Document Sections

- Introduction
- MMA-CNN
- Experimental Results
- Conclusion

**Abstract:**

To apply convolutional neural networks (CNNs) on high-resolution images, a common approach is to split the input image into smaller patches. However, the field-of-view is restricted by the input size. To overcome the problem, a multi-magnification attention convolutional neural network (MMA-CNN) is proposed to analyze images based on both local and global features. Our approach focuses on identifying the importance of individual features at each magnification level and is applied to pathology whole slide images (WSIs) segmentation to show its effectiveness. Several interactive figures are also developed to enhance the reader's understanding of our research.

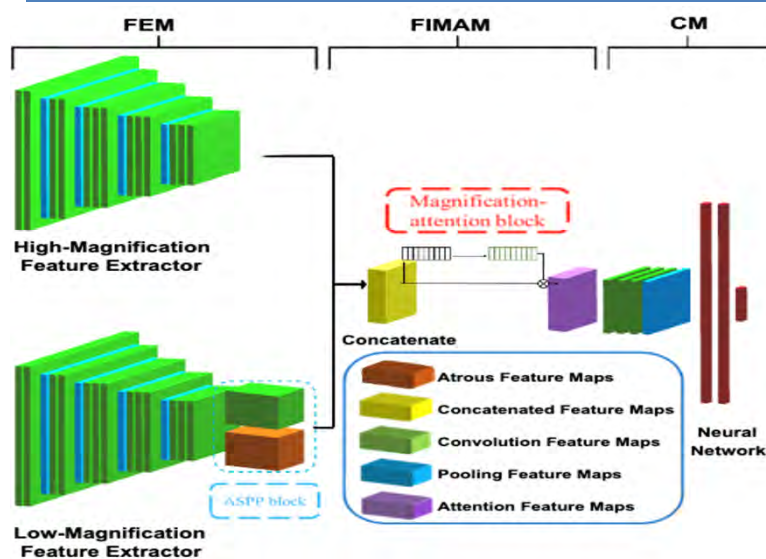
**Published in:** IEEE Computational Intelligence Magazine ( Volume: 18 , Issue: 3, August 2023)

**Page(s):** 54 - 55

**INSPEC Accession Number:** 23452917

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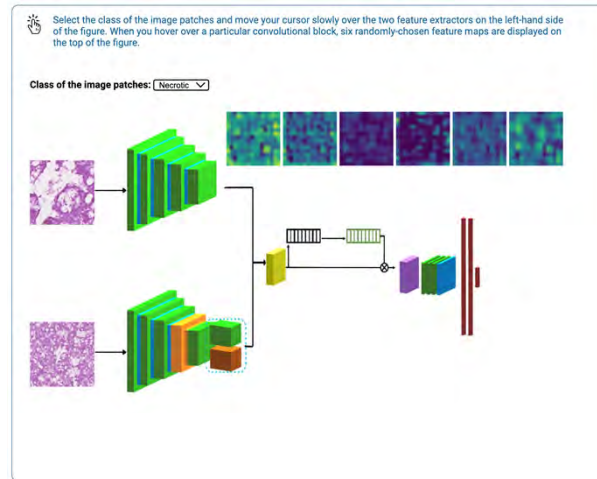
## Multi-Magnification Attention Convolutional Neural Networks



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## Feature Extraction Module

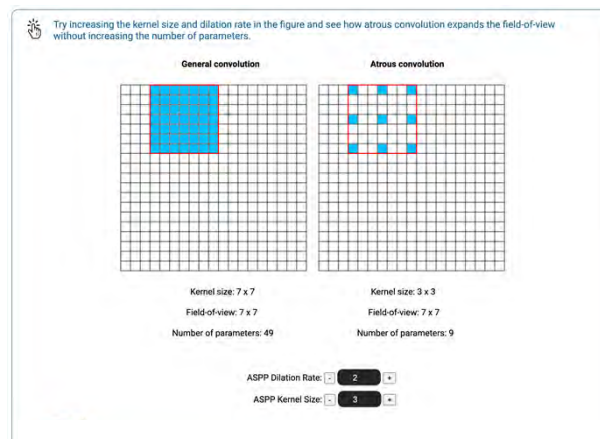
- Extract the feature maps at various magnification



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## Feature Extraction Module

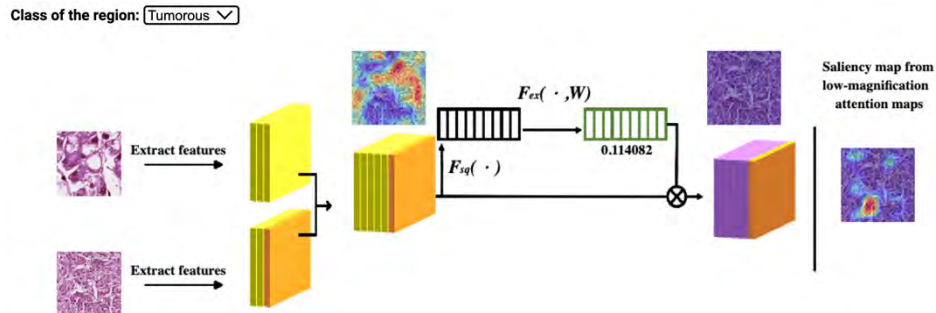
- Expand the field-of-view with atrous spatial pyramid pooling (ASPP)



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## Feature Integration and Magnification Attention Module

- Integrate features from images of different magnification factors



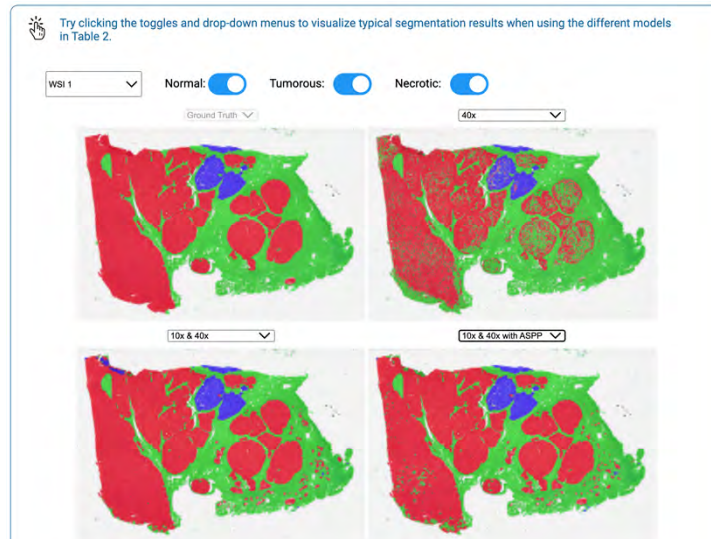
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## HCC Dataset

| Network Structure | Metric Scale | Tumor    |             |       | Necrosis    |       |
|-------------------|--------------|----------|-------------|-------|-------------|-------|
|                   |              | Accuracy | Sensitivity | IoU   | Sensitivity | IoU   |
| Single Network    | 40x          | 0.955    | 0.890       | 0.846 | 0.935       | 0.926 |
| Single Network    | 10x          | 0.964    | 0.923       | 0.887 | 0.964       | 0.951 |
| Single Network    | 2.5x         | 0.970    | 0.938       | 0.894 | 0.959       | 0.947 |
| Parallel Networks | 40x&10x      | 0.980    | 0.968       | 0.936 | 0.967       | 0.950 |
| Parallel Networks | 10x&2.5x     | 0.974    | 0.985       | 0.913 | 0.973       | 0.954 |
| MMA-CNN           | 40x&10x      | 0.986    | 0.991       | 0.977 | 0.967       | 0.958 |
| MMA-CNN           | 10x&2.5x     | 0.980    | 0.976       | 0.937 | 0.977       | 0.955 |

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## Visualization



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## Transformer-Based Approach

### Challenge: Fewer Pathology Images

Compared with the natural image datasets, the number of pathology images is significantly fewer and boundaries of the tissues are relatively unclear.

### Research Question

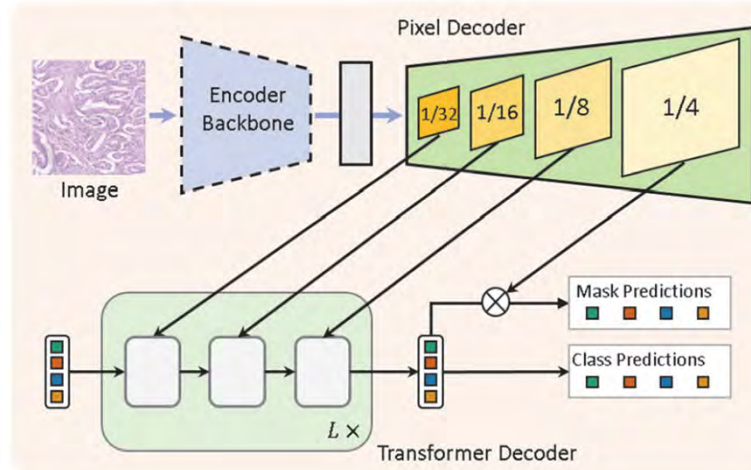
We want to explore whether the mainstream transformer-based method is also applicable to solve the instance segmentation problem of the pathology datasets.

### Solution: Mask2Former

We use Mask2Former to solve the pathology image instance segmentation problem, leveraging its powerful masked attention mechanism for precise gland delineation.

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## Mask2Former



<https://arxiv.org/abs/2112.01527>

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## Results of Glas Dataset

| Methods         |     | F1    | Dice  | Hausdorff |
|-----------------|-----|-------|-------|-----------|
| Mask R-CNN [3]  | (A) | 0.779 | 0.815 | 80.19     |
|                 | (B) | 0.578 | 0.679 | 174.56    |
| CMD-Net [19]    | (A) | 0.919 | 0.912 | 40.13     |
|                 | (B) | 0.860 | 0.848 | 98.32     |
| GCSBA-Net [20]  | (A) | 0.916 | 0.914 | 41.49     |
|                 | (B) | 0.832 | 0.834 | 102.88    |
| HFE-Net [21]    | (A) | 0.923 | 0.924 | 34.89     |
|                 | (B) | 0.801 | 0.858 | 86.09     |
| Mask2Former [1] | (A) | 0.905 | 0.901 | 48.63     |
|                 | (B) | 0.813 | 0.842 | 95.06     |

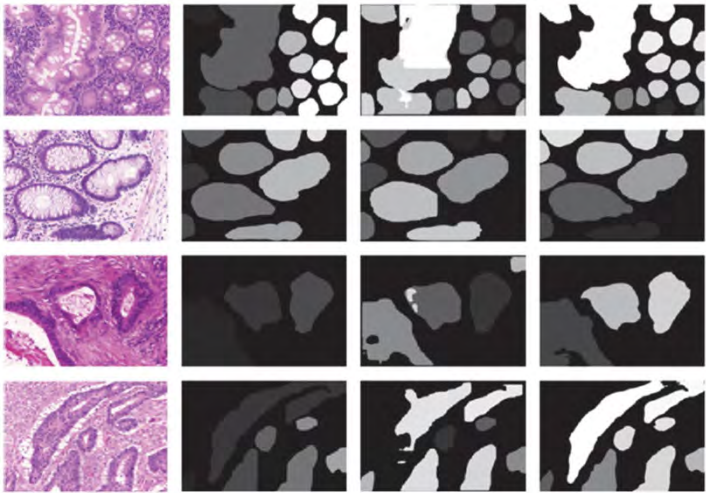
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## Results of Crag Dataset

| Methods         | F1    | Dice  | Hausdorff |
|-----------------|-------|-------|-----------|
| Mask R-CNN [3]  | 0.771 | 0.799 | 190.99    |
| CMD-Net [19]    | 0.840 | 0.879 | 132.38    |
| GCSBA-Net [20]  | 0.836 | 0.894 | 146.77    |
| HFE-Net [21]    | 0.820 | 0.887 | 110.05    |
| Mask2Former [1] | 0.847 | 0.899 | 104.70    |

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## GlaS Dataset

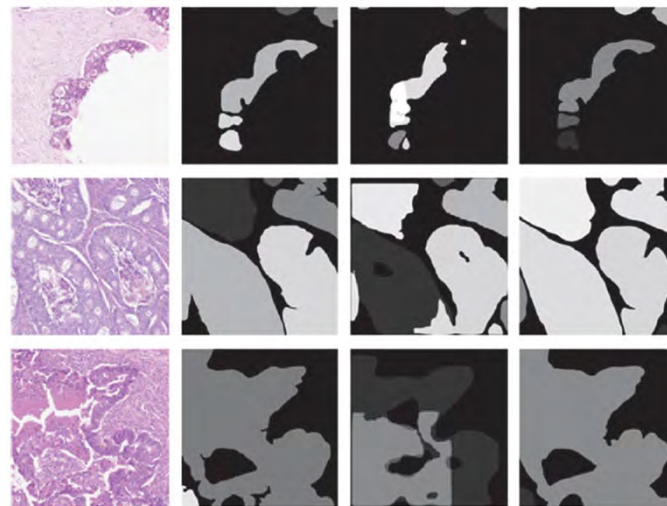


Original Images      GT      Mask-RCNN      Mask2Former

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## CRAG Dataset

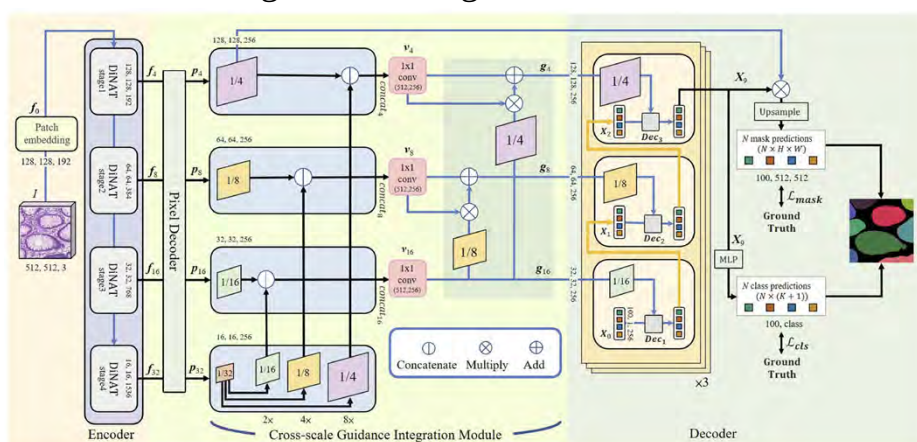


Original Images GT Mask-RCNN Mask2Former

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## More Advanced Network

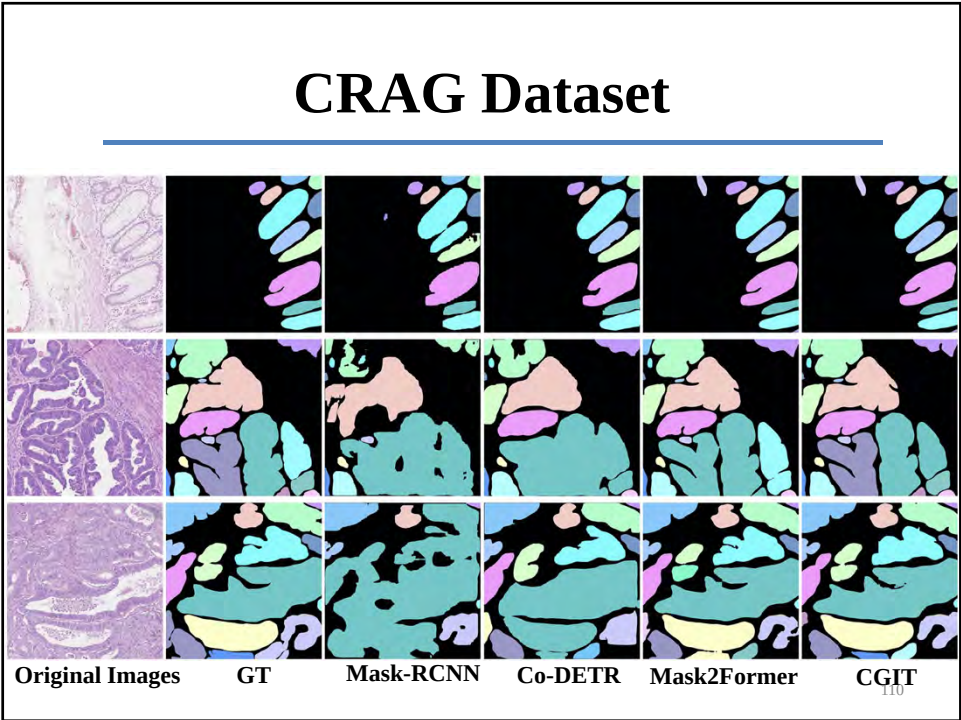
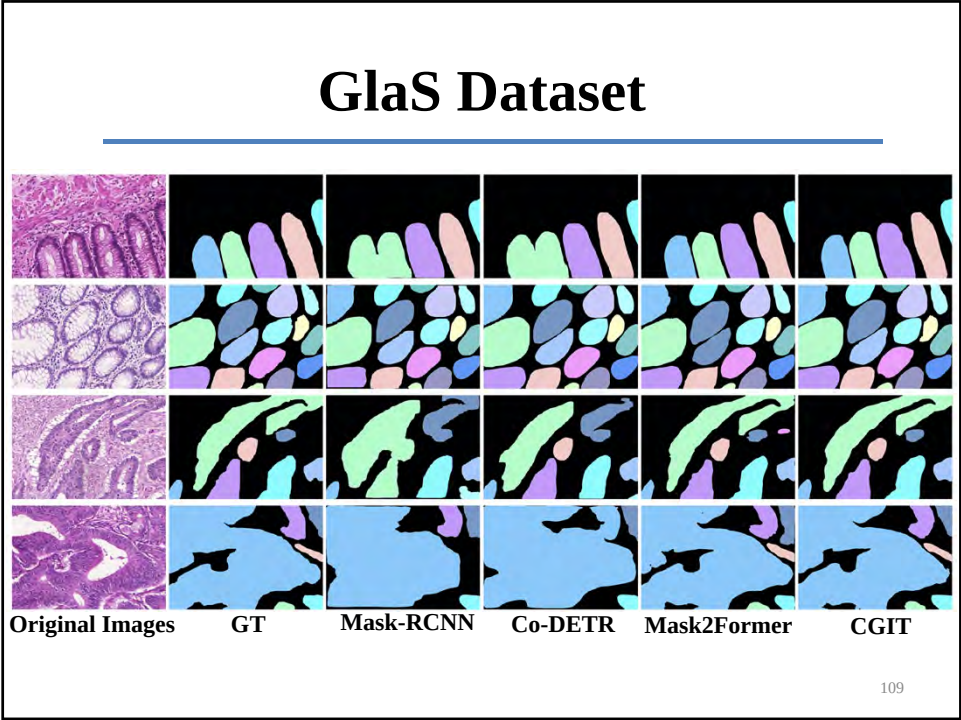
- Cross-scale guidance integration transformer



<https://ieeexplore.ieee.org/document/10945390/>

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## Problem of Supervised Learning?

### Labeling is Extremely Time-Consuming

Due to the extremely high resolution of WSIs and ambiguous boundaries between regions of tumor and normal liver tissues, the manual labeling process is very time-consuming.

### Key Research Challenge

Thus, how to train good deep segmentation models by using fewer labeled training data becomes an important issue in pathological WSI analysis.

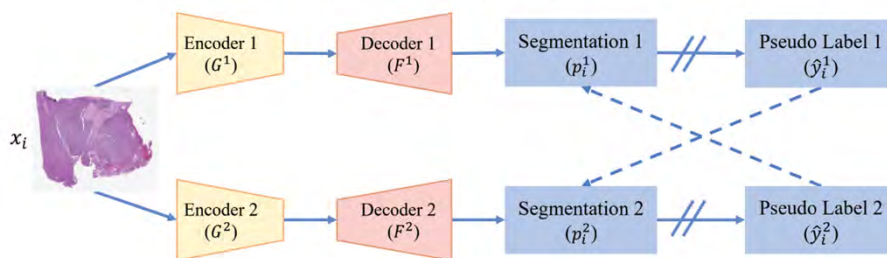
### Semi-supervised Solution

To address this issue, semi-supervised segmentation methods are proposed to leverage unlabeled training data and reduce the need for labeled data.

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## Semi-supervised Segmentation Method

- Cross pseudo supervision (CPS)



#### Network A

First segmentation network trained on labeled data and generates pseudo-labels for unlabeled samples to supervise Network B.

#### Network B

Second segmentation network with different initialization, generating pseudo-labels to supervise Network A in return.

#### Cross Supervision

The mutual pseudo-label supervision between the two networks effectively leverages unlabeled data to improve segmentation performance.

<https://arxiv.org/abs/2106.01226>

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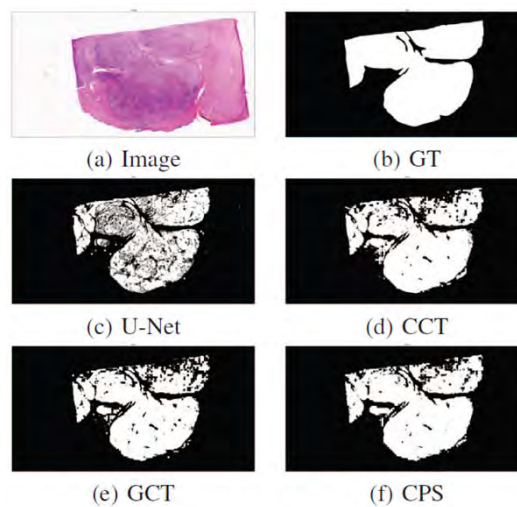
## Semi-supervised Tumor Segmentation Results

| Method    | Param. (M) | FLOPS (G) | 20L 60U | 30L 50U | 40L 40U | 80L    |
|-----------|------------|-----------|---------|---------|---------|--------|
| U-Net [2] | 13.16      | 5.39      | -       | -       | -       | 83.80% |
| CCT [4]   | 40.48      | 19.13     | 81.26%  | 79.48%  | 80.03%  | -      |
| GCT [5]   | 89.22      | 19.13     | 79.55%  | 80.06%  | 82.37%  | -      |
| CPS [6]   | 26.32      | 5.39      | 83.48%  | 84.42%  | 84.55%  | -      |

- **L: Labeled data, U: Unlabeled data** — Results show that incorporating unlabeled data through semi-supervised learning significantly improves segmentation accuracy compared to using labeled data alone.

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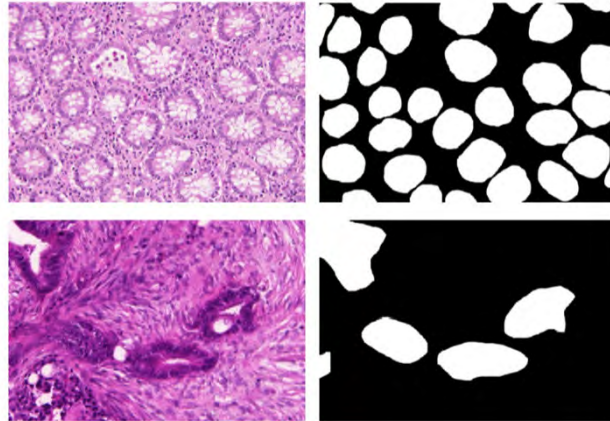
## Tumor Segmentation Results



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## Cell Segmentation

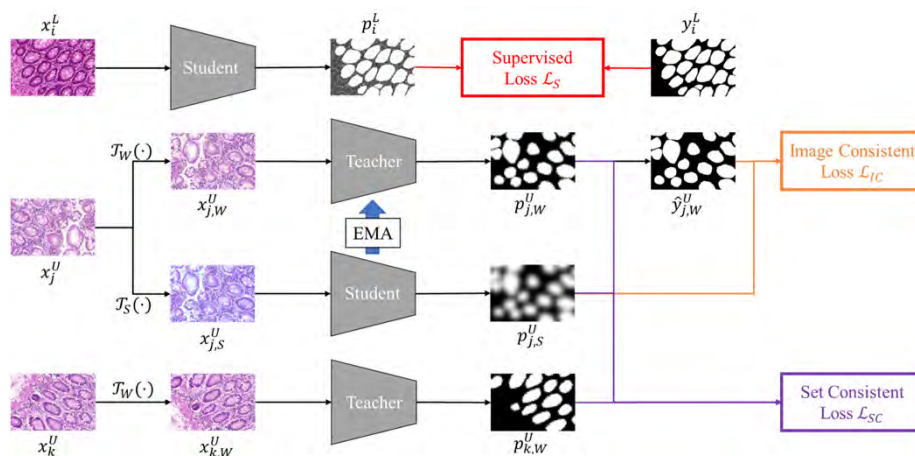
- Identify cells from pathology images
- Time consuming labeling



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## Reducing Annotations

- Student/Teacher Models

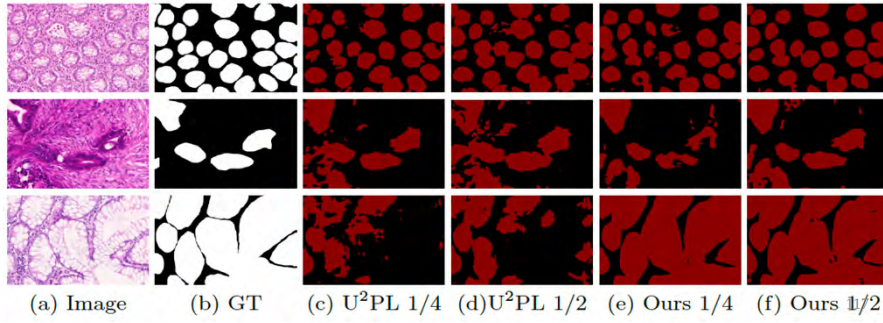


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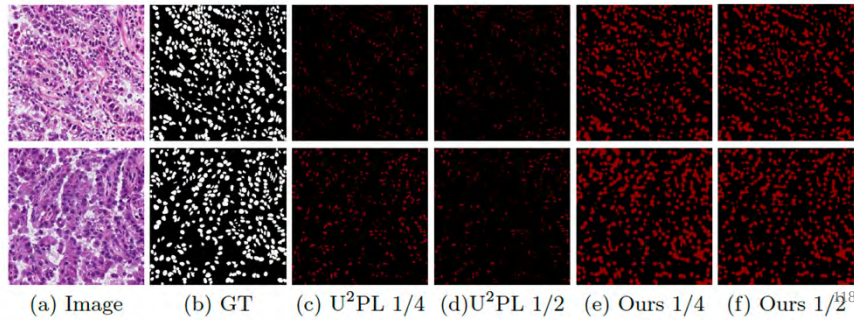
## GlaS Dataset

| Method                 | Backbone       | 1/16  | 1/8   | 1/4   | 1/2   | 1     |
|------------------------|----------------|-------|-------|-------|-------|-------|
| U-Net [21]             | U-Net          | -     | -     | -     | -     | 76.81 |
| Attention U-Net [19]   | AttentionU-Net | -     | -     | -     | -     | 77.53 |
| TransU-Net [5]         | TransU-Net     | -     | -     | -     | -     | 79.10 |
| MedT [26]              | MedT           | -     | -     | -     | -     | 77.50 |
| SwinU-Net [3]          | SwinU-Net      | -     | -     | -     | -     | 79.86 |
| NMNet [24]             | NMNet          | -     | -     | -     | -     | 82.13 |
| ReCo [17]              | ResNet-101     | 61.68 | 62.84 | 64.59 | 66.57 | -     |
| U <sup>2</sup> PL [27] | ResNet-101     | 29.74 | 41.14 | 63.26 | 70.22 | -     |
| Proposed               | ResNet-50      | 68.98 | 69.83 | 78.18 | 83.37 | -     |



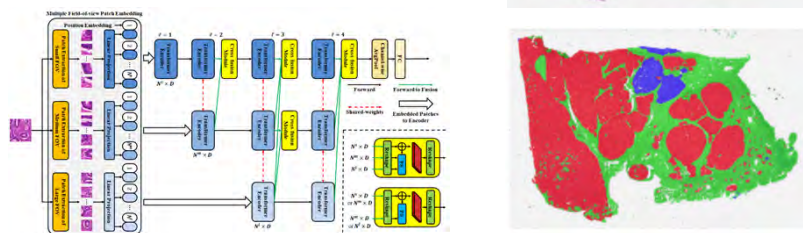
## MoNuSeg Dataset

| Method                 | Backbone       | 1/16  | 1/8   | 1/4   | 1/2   | 1     |
|------------------------|----------------|-------|-------|-------|-------|-------|
| U-Net [21]             | U-Net          | -     | -     | -     | -     | 59.81 |
| Attention U-Net [19]   | AttentionU-Net | -     | -     | -     | -     | 62.64 |
| TransU-Net [5]         | TransU-Net     | -     | -     | -     | -     | 65.68 |
| MedT [26]              | MedT           | -     | -     | -     | -     | 65.73 |
| SwinU-Net [3]          | SwinU-Net      | -     | -     | -     | -     | 64.72 |
| NMNet [24]             | NMNet          | -     | -     | -     | -     | 67.73 |
| ReCo [17]              | ResNet-101     | 51.75 | 69.26 | 72.62 | 75.68 | -     |
| U <sup>2</sup> PL [27] | ResNet-101     | 58.86 | 69.04 | 64.59 | 69.77 | -     |
| Proposed               | ResNet-50      | 71.51 | 72.26 | 70.33 | 73.50 | -     |



## Summary

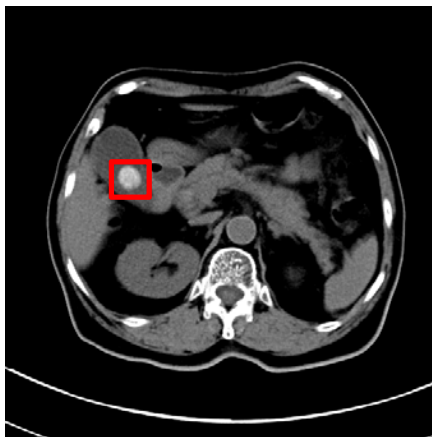
- Deep learning in pathology image analysis
  - Classification
  - Segmentation
  - ...
- General AI models ?



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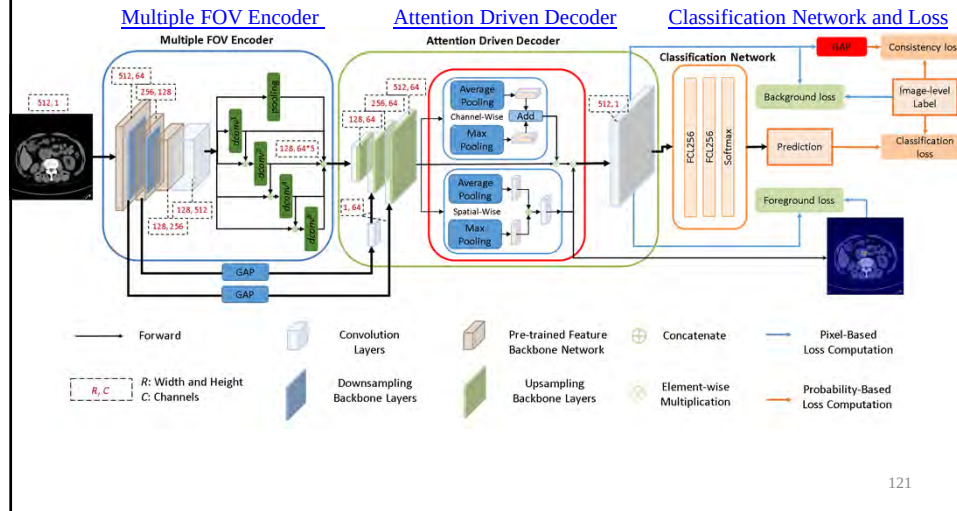
## CT Image Analysis

- Gallstones vs. Common bile duct (CBD)



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## Multiple Field-Of-View Based Attention Driven Network



## Quantitative Results

| Methods  | Accuracy      | Sensitivity   | Specificity   | F1-score      |
|----------|---------------|---------------|---------------|---------------|
| CAM      | 0.8179        | 0.8077        | 0.8231        | 0.7473        |
| Grad-CAM | 0.8128        | 0.7308        | 0.8538        | 0.7224        |
| MS-CAM   | 0.8154        | 0.8462        | 0.8000        | 0.7534        |
| SPA      | 0.8846        | 0.8077        | <b>0.9231</b> | 0.8235        |
| Ours     | <b>0.9051</b> | <b>0.8692</b> | <b>0.9231</b> | <b>0.8593</b> |

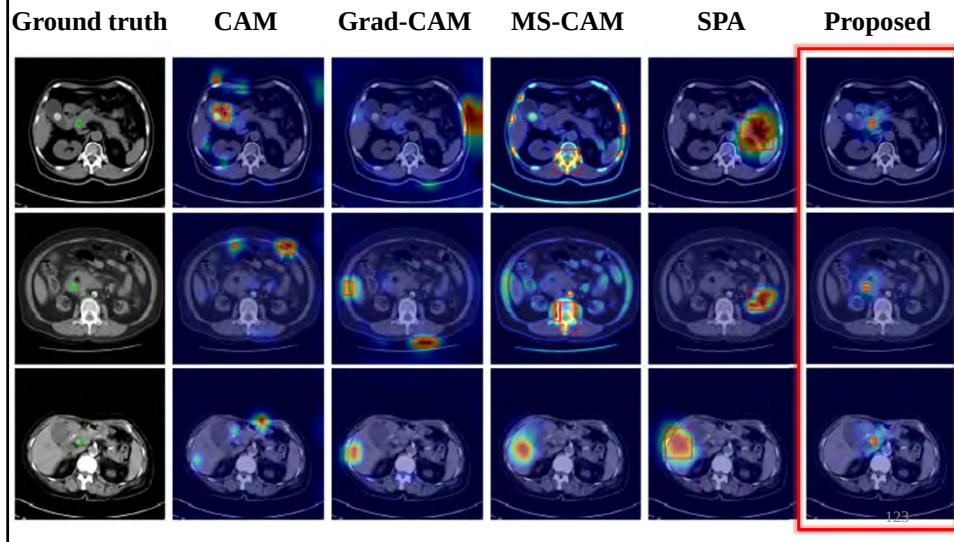
| Methods  | mIoU          | AP            |
|----------|---------------|---------------|
| CAM      | 0.0206        | 0.0263        |
| Grad-CAM | 0.0011        | 0.0002        |
| MS-CAM   | 0.0127        | 0.0108        |
| SPA      | 0.0015        | 0.0017        |
| Ours     | <b>0.5309</b> | <b>0.5707</b> |

Quantitative results confirm that the proposed Multiple FOV Attention Driven Network achieves state-of-the-art performance in distinguishing gallstones from CBD stones on CT images.

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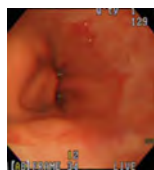
## Visualizations



## Conclusions

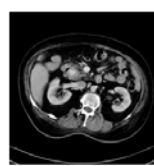
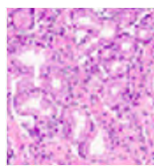
### Deep Learning Beyond Natural Images

Deep learning is not only powerful for natural images, but also demonstrates exceptional capability in medical image analysis across diverse modalities



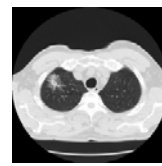
### Medical Image Applications

- **Endoscopic images** — gastrointestinal lesion detection
- **Pathology images** — tissue classification and segmentation
- **CT images** — stone detection and organ analysis
- ... and many more emerging modalities



### More Medical Applications

The frontier of medical AI continues to expand — from general-purpose foundation models to specialized clinical decision support systems



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**Thank you for your  
listening**

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